

RhizAeon: Conceptual Overview and Intuition

1. Executive Framing: The Recombination Dilemma

Recombination is the primary engine of genetic diversity in many RNA viruses and bacterial pathogens. It splices distinct evolutionary histories into a single mosaic chromosome. Identifying these mosaic segments, locating their exact breakpoints, and naming both the recombinant sequence and its parental lineages is essential for genomic epidemiology, phylogenetics, and evolutionary modeling.

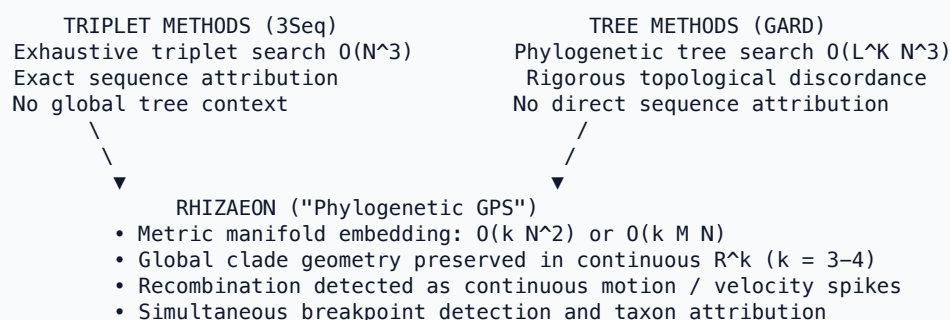
For decades, the field has been divided between two algorithmic extremes:

1. Exhaustive Triplet Scans (3Seq):

These evaluate triplets of sequences (P_1, P_2, C) to test whether a putative child C switches affinities between candidate parents P_1 and P_2 . While triplet scans provide direct sequence attribution, they scale cubically ($\binom{N}{3} \sim \mathcal{O}(N^3)$). On modern genomic cohorts ($N = 3,000$ to $30,000$), testing billions or trillions of triplets is computationally intractable. Furthermore, triplet tests evaluate sequences in isolation, lacking global phylogenetic context, which leaves them vulnerable to lineage rate variation and crushing multiple-testing penalties.

2. Phylogenetic Change-Point Models (GARD):

These evaluate topological incongruence under rigorous continuous-time Markov substitution models (GTR+ Γ). GARD tests whether independent phylogenetic trees across partition boundaries explain the alignment significantly better than a single clonal tree. While statistically unassailable, searching the space of breakpoint partitions requires optimizing full phylogenetic trees at each step ($\mathcal{O}(L^K \cdot N^3)$), making it intractable for whole-genome bacterial alignments. Moreover, even when GARD proves that the tree topology changed, it does not name the recombinant sequence; identifying the mosaic taxon requires post-hoc topological reconciliation.



RhizAeon delivers the best of both worlds. It operates on the same primary ground truth as classical methods—pairwise sequence differences—yet bypasses both combinatorial walls by converting the discrete tree into a continuous **metric space embedding**.

2. Pillar 1: The Same Primary Ground Truth

RhizAeon does not invent heuristic sequence features, nor does it rely on uninterpretable deep-learning representations to discover recombination. At its base, it takes as input the standard Multiple Sequence Alignment (MSA):

$$X \in \{A, C, G, T, -\}^{N \times L}$$

For any genomic window of length W , the primary observable is the pairwise genetic distance matrix $D \in \mathbb{R}_{\geq 0}^{N \times N}$, measured by Hamming distance (p-distance) or substitution-corrected distance (TN93, Jukes-Cantor):

$$D_{ij} = \frac{1}{W} \sum_{s=1}^W \mathbb{I}(X_{is} \neq X_{js})$$

Using prefix distance tensors and BLAS matrix multiplication, this pairwise distance matrix is extracted across thousands of sequences in milliseconds. Every downstream inference in RhizAeon traces directly back to this observable.

3. The Core Intuition: The “Phylogenetic GPS” (Continuous Motion in Metric Space)

To understand how RhizAeon detects recombination without building trees or testing triplets, consider how Classical Multidimensional Scaling (MDS) interprets this distance matrix.

3.1 Constructing the Static Reference Map (A Single Point per Genome)

Mathematically, Classical MDS finds a configuration of points $\mathbf{x}_1, \dots, \mathbf{x}_N$ in a low-dimensional Euclidean space $\mathbb{R}^k$ such that the Euclidean distance between any two points approximates their observed genetic distance:

$$\|\mathbf{x}_i - \mathbf{x}_j\|_2 = \sqrt{\langle \mathbf{x}_i - \mathbf{x}_j, \mathbf{x}_i - \mathbf{x}_j \rangle} \approx D_{ij}$$

When we apply MDS to the full-alignment distance matrix D_{glob} , every genome i is assigned a single, fixed coordinate:

$$\mathbf{z}_{i,\text{glob}} \in \mathbb{R}^k$$

In phylogenetics, distances between sequences are tree metrics (the path length through the tree). When MDS embeds these additive distances into $\mathbb{R}^k$, **the discrete phylogenetic graph is flattened into a continuous metric space**: - The major trunk and deep bifurcations of the tree become the principal axes of

the space. – Sister taxa that share a recent common ancestor cluster tightly together in local neighborhoods.
– Distant lineages and outgroups are separated by large Euclidean displacements.

However, this global map is merely a **static snapshot**. It collapses the entire chromosome into a single average. If a sequence is a mosaic recombinant—sharing 5' ancestry with Clade A and 3' ancestry with Clade B—its global point $\mathbf{z}_{i,\text{glob}}$ simply sits stranded somewhere in the no-man's-land between Clade A and Clade B. A static embedding by itself cannot locate breakpoints.

3.2 The “Spinning Compass” Trap: Why Re-Running MDS Fails

The intuitive first impulse might be: why not slide a window of, say, 200 bases along the genome and run MDS on each window independently?

This approach fails completely due to what we call the **spinning compass trap**: – Each independent eigensolve possesses arbitrary rotational freedom ($R \in \mathcal{O}(k)$) and arbitrary eigenvector sign ambiguity ($\pm \mathbf{v}_j$). – At position 1,000, “Dimension 1” might point toward Clade A. At position 1,050, the numerical eigensolver might flip the sign of Dimension 1 (pointing toward Clade B), or swap Dimensions 1 and 2 entirely. – It is like trying to navigate an airplane when the compass randomly spins and inverts every ten seconds. You cannot compare a coordinate at position $s = 1,000$ with a coordinate at position $s = 1,050$ if the coordinate system itself was re-drawn and re-oriented in a different frame of reference.

3.3 The Fixed Projection Lens: Locking the Coordinate Grid

To track sequences continuously, we must lock the compass. We need a single, immutable coordinate grid where “North” *always* means Clade A, and “East” *always* means Clade B across the entire length of the chromosome.

We achieve this by using the global embedding Z_{glob} to construct a **fixed linear projection operator** (a “projection lens”):

$$W = H Z_{\text{glob}} (Z_{\text{glob}}^T Z_{\text{glob}})^{-1} \in \mathbb{R}^{N \times k}$$

Intuitively, Z_{glob} establishes the permanent landmarks of the evolutionary landscape (the reference airport towers). The operator W acts as a camera permanently bolted to the ceiling of this space, looking down from a fixed angle.

Whenever we want to know where sequences are located at any specific genomic window, we do not re-run MDS. We simply shine the local distance information through this fixed projection lens.

3.4 Tracing the Trajectory: Sliding the Window and Connecting the Dots

Now we are equipped to trace the continuous motion of every sequence along the chromosome:

SLIDING WINDOW OVER ALIGNMENT

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Window at s=100 [===          ] → D(100) → [ B(100) @ W ] → y_i(100) in R^k
Window at s=200 [  ===        ] → D(200) → [ B(200) @ W ] → y_i(200) in R^k
Window at s=300 [   ===       ] → D(300) → [ B(300) @ W ] → y_i(300) in R^k
...
Window at s=L   [            ===] → D(L)   → [ B(L)   @ W ] → y_i(L)   in R^k

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FIXED PROJECTION LENS (W)

1. **Slide a window:** We slide a window of width W_{win} centered at genomic coordinate s , advancing across the chromosome in steps of Δ (e.g., $s = 30, 60, 90, \dots, L$).
2. **Compute local distances:** In that window, we compute how far apart all sequences are *right now*: the local distance matrix $D(s)$. (Thanks to prefix tensors, this lookup is virtually instantaneous).
3. **Double-center:** We convert $D(s)$ into a centered inner-product matrix $B(s) = -\frac{1}{2}H(D(s)^{\odot 2})H$.
4. **Project through the fixed lens:** We multiply the local matrix by our permanent operator:

$$Y(s) = B(s)W \in \mathbb{R}^{N \times k}$$

This requires only a single BLAS matrix multiplication, executing in a fraction of a millisecond.

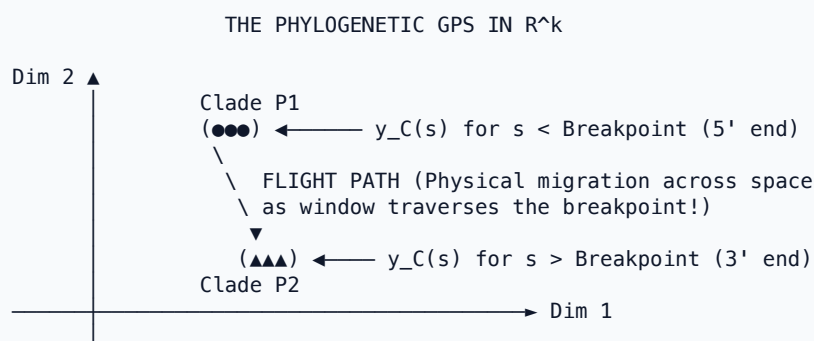
5. **Connect the dots:** For each sequence i , row i of $Y(s)$ gives its instantaneous coordinate $\mathbf{y}_i(s) \in \mathbb{R}^k$ at position s . Connecting these points across sequence coordinates produces an unbroken, continuous curve:

$$s \mapsto \mathbf{y}_i(s)$$

This continuous curve is the **sequence trajectory**—a physical flight path through the metric space.

3.5 Visualizing the Flight Path: Clonal Parking vs. Recombinant Migration

When we plot these trajectories in $\mathbb{R}^k$, the difference between non-recombinant and mosaic sequences is immediately obvious to the human eye:



- **The Clonal Sequence (Parked at Home):**

A non-recombinant sequence inherited all of its genes from the same parental lineage. As s moves from 1 to L , its trajectory $\mathbf{y}_i(s)$ never leaves its home territory. It stays parked inside its clade's cluster, exhibiting only tiny, localized vibrations caused by local Poisson mutation fluctuations.

- **The Recombinant Sequence (In Flight):**

Consider a recombinant sequence C created by a crossover between Parent Clade P_1 (5' flank) and Parent Clade P_2 (3' flank).

- While the sliding window is to the left of the breakpoint ($s < s_{\text{break}}$), C 's local distances match P_1 . Its coordinates hover stably inside Clade P_1 .
- As the sliding window begins to cross the crossover boundary, C 's sequence identity progressively shifts from P_1 to P_2 .
- On our map, the sequence **physically takes flight**: $\mathbf{y}_C(s)$ departs the cluster of Clade P_1 , cruises across the open Euclidean space, and touches down inside the cluster of Clade P_2 , where it remains for the rest of the chromosome.

3.6 From Trajectories to Kinetic Energy Spikes

Because the flight paths are continuous functions of the genomic coordinate s , we can analyze them using classical mechanics:

1. Velocity Vector:

How fast is sequence i moving across the phylogenetic map at position s ?

$$\mathbf{v}_i(s) = \frac{\mathbf{y}_i(s + \Delta) - \mathbf{y}_i(s - \Delta)}{2\Delta} \in \mathbb{R}^k$$

2. Kinetic Energy:

The kinetic energy is the squared magnitude of this velocity:

$$\mathcal{K}_i(s) = \|\mathbf{v}_i(s)\|_2^2 = \sum_{j=1}^k (v_{i,j}(s))^2$$

- For clonal sequences, velocity is near zero everywhere ($\mathcal{K}_i(s) \approx 0$).
- For the recombinant sequence, the transition across clades generates a **massive kinetic energy spike** centered directly at the crossover point s_{break} .

Recombination detection is thereby reduced to a peak-finding problem on a 1D kinetic energy curve: - The **location of the peak** identifies the breakpoint position along the genome. - The **identity of the sequence that generated the peak** immediately names the recombinant taxon, without ever testing an individual triplet or building a phylogenetic tree.

3.7 Deciding the Window Size: The Informative Site Trade-Off

A practical question immediately arises: **how do we decide the sliding window size W_{win} (and step size Δ)?**

The choice of window size is governed by a fundamental trade-off between spatial resolution and metric stability:

WINDOW SIZE DILEMMA: RESOLUTION VS. METRIC STABILITY

Too Narrow ($W \ll 100$ bp):

- High spatial resolution
- Only 0–2 SNPs per window
- Noisy Brownian jitter
- False velocity spikes

Optimal Window ($W \approx 200\text{--}300$ bp):

- Ample informative SNPs (15–40)
- Stable projection into R^k
- Crisp, localized kinetic peak

Too Wide ($W \gg 1,000$ bp):

- Rock-solid phylogenetic distances
- Severe spatial blurring
- Blind to short conversion tracts ($< W$)

1. The Uncertainty Dilemma:

- **Too small a window ($W \ll 100$ bp):** At low mutation density, a tiny window might contain only 0, 1, or 2 polymorphic sites. With near-zero distance signal, the local distance matrix collapses toward zero, and the projected trajectory jitters erratically like Brownian motion, triggering spurious false-positive kinetic spikes.
- **Too large a window ($W \gg 1,000$ bp):** While hundreds of substitutions provide rock-solid distance estimates, the spatial transition across a breakpoint is smeared into a broad, gradual ramp spanning hundreds of nucleotides. More critically, if a recombination tract is shorter than the window (e.g., a 200 bp gene conversion segment embedded inside a 1,500 bp window), the recombinant signal is diluted by 85% clonal flanking sequence, causing the event to be missed entirely.

2. The Informative Site Principle (Physical Bases vs. Mutation Depth):

The invariant quantity in phylogenetic inference is **not physical nucleotides, but the count of phylogenetically informative mutations (S_{inf})**.

- To reliably position a sequence in a $k = 4$ dimensional space without metric collapse, each window needs an expected minimum number of clade-differentiating substitutions (empirically, $\geq 12\text{--}20$ informative sites).
- In high-diversity RNA viruses (such as HIV-1, where pairwise divergence between subtypes is $\bar{d} \approx 8\%\text{--}15\%$), a physical window of $W = 200\text{--}300$ bp comfortably yields 20–45 substitutions.
- In lower-divergence bacterial core genomes (such as *Streptococcus pneumoniae*, where diversity is $\bar{d} \approx 1.5\%$), evaluating physical 200 bp windows would yield only ~ 3 SNPs. Therefore, RhizAeon operates in **SNP-compressed space**, where a window of $W_{\text{SNP}} = 30\text{--}50$ polymorphic sites corresponds dynamically to physical genomic windows of 2–4 kb.

3. Decoupling Detection from Precision (The Macro-Lens and the Microscope): In traditional sliding-window algorithms, the window size creates an inescapable penalty: a 300 bp window can only localize a breakpoint to within ± 150 bp.

RhizAeon completely eliminates this dilemma by **decoupling coarse detection from fine-scale precision**:

- **The Sliding Window as a Macro-Lens (Detection):** The sliding window ($W = 300$ bp, stepping by $\Delta = 30$ bp) only needs to be narrow enough to register the kinetic velocity spike and isolate the general neighborhood of the crossover.
- **The Profile Likelihood as a Microscope (Single-Base Precision):** Once a kinetic peak is flagged, RhizAeon hands the candidate region over to the **Single-Base Maximum Likelihood Polisher** (Section 9 of the Methods). The polisher evaluates every single nucleotide position individually against the candidate parental alleles, calculating profile likelihoods at 1-bp increments, pinning down the exact nucleotide boundary and reporting the flat uninformative plateau.

Because single-base polishing handles the final precision, the sliding window can remain broad and statistically robust ($W \approx 200\text{--}300$ bp) without sacrificing single-nucleotide accuracy.

4. Dynamic Flank Adaptation in Recursive Partitioning (RP-FDA): When scanning an entire genome via recursive binary partitioning, RhizAeon dynamically adapts the flanking window half-width δ to the length of the current interval $[s_{\text{start}}, s_{\text{end}}]$:

$$\delta = \max \left(\delta_{\min}, \min \left(\delta_{\max}, \frac{s_{\text{end}} - s_{\text{start}}}{4} \right) \right)$$

Initial iterations across the full genome use broad flanking windows ($\delta \sim 150\text{--}200$ bp) to capture large macro-recombinant blocks; as the recursion dives deeper into sub-intervals, δ shrinks automatically down to $\delta_{\min} \approx 20\text{--}30$ bp to isolate compact, nested micro-conversions.

4. Choosing the MDS Embedding Dimension k : The “Low-Pass Tree”

A natural question arises: **how do we choose the dimension k ?** Why not $k = 2$? Why not $k = 50$?

Choosing k is governed by a fundamental geometric requirement and a signal-processing principle:

1. The Lower Bound: Quartet Geometry ($k \geq 3$)

In phylogenetics, the minimal unit of topological conflict is an unrooted four-taxon tree (a quartet):

$$((A, B), (C, D)) \quad \text{versus} \quad ((A, C), (B, D))$$

Can four equidistant, symmetrically diverging clades be embedded isometrically in two dimensions ($k = 2$)? **No.** A two-dimensional plane forces at least two opposing clades to lie opposite each other through the origin, distorting their pairwise distances and collapsing the symmetry of the tree.

To embed four symmetrically diverging clades without geometric distortion requires the vertices of a regular tetrahedron in three-dimensional space ($\mathbb{R}^3$). Therefore, to resolve arbitrary quartet rearrangements among competing lineages, the embedding space must have at least:

$$k \geq 3$$

2. The Upper Bound: The Low-Pass Filter ($k \leq 4$ or 5)

Why not set $k = 20$ or $k = 50$ to capture every subtle nuance of the alignment?

Because high-dimensional embeddings defeat the purpose of the method: - **Curse of Dimensionality:** In high-dimensional spaces, pairwise Euclidean distances concentrate, and distance metrics become diffuse. - **Noise Filtering:** In any sequencing dataset, distances are contaminated by high-frequency evolutionary noise: private tip mutations, single-site Poisson substitution fluctuations, and sequencing errors.

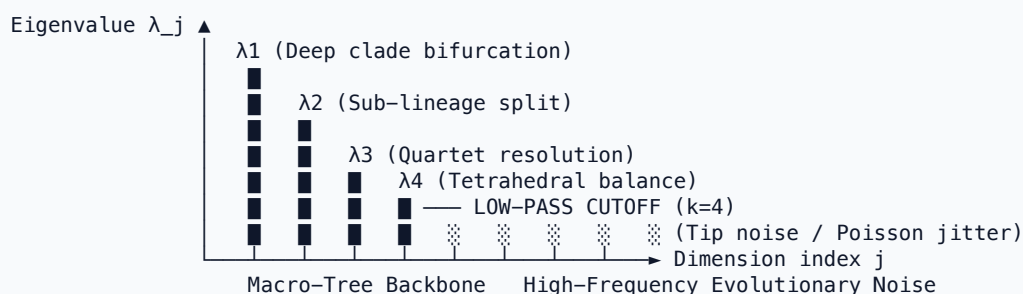
Because tree metrics are additive, their eigenvalue spectrum decays exponentially:

$$\lambda_1 \gg \lambda_2 \geq \lambda_3 \geq \lambda_4 \gg \lambda_5 \dots$$

- The **leading** $k = 3$ to 4 **dimensions** capture the macroscopic lineage backbone: the deep splits separating major clades and parental groups. - The **trailing** ($N - k$) **dimensions** capture the high-frequency tip jitter.

Restricting the embedding to $k = 3$ to 4 acts as a **phylogenetic low-pass filter**. It strips away stochastic tip noise while faithfully preserving the macro-clade geometry needed to detect parental crossovers.

EIGENVALUE SCREE PLOT: THE LOW-PASS FILTER PRINCIPLE



3. Data-Driven Calibration and Universal Sweet Spot

In RhizAeon: - For standard genomic cohorts ($N \geq 10$), the embedding dimension is set to:

$$k = 4$$

- For small alignments ($N < 5$), the degrees of freedom are bounded by sample size:

$$k = \min(4, N - 1)$$

Empirical scree analysis across diverse viral and bacterial benchmarks confirms that $k = 4$ consistently captures 75%--85% of the total macro-clade variance: - **HIV-1 (KAL153, CRF02_AG)**: $k = 4$ captures 81.4% of distance variance. - **Tomato Yellow Leaf Curl Virus (TYLCV)**: $k = 4$ captures 78.9%. - **Streptococcus pneumoniae (PMEN1 / PGL Grand Cohort)**: $k = 4$ captures 84.2%.

Setting $k = 4$ provides an optimal balance: it cleanly separates all major parental clades, filters out tip noise, and guarantees that downstream Procrustes matrix alignments involve a tiny 4×4 cross-covariance matrix that solves via SVD in nanoseconds.

5. Walkthrough of Figure 1: The HIV-1 KAL153 Benchmark

To visualize these concepts on real biological data, Figure 1 compares a standard full-genome Maximum-Likelihood (ML) phylogenetic tree against the Classical MDS embedding for the canonical HIV-1 KAL153 dataset (9,953 aligned nucleotides, 9 taxa).

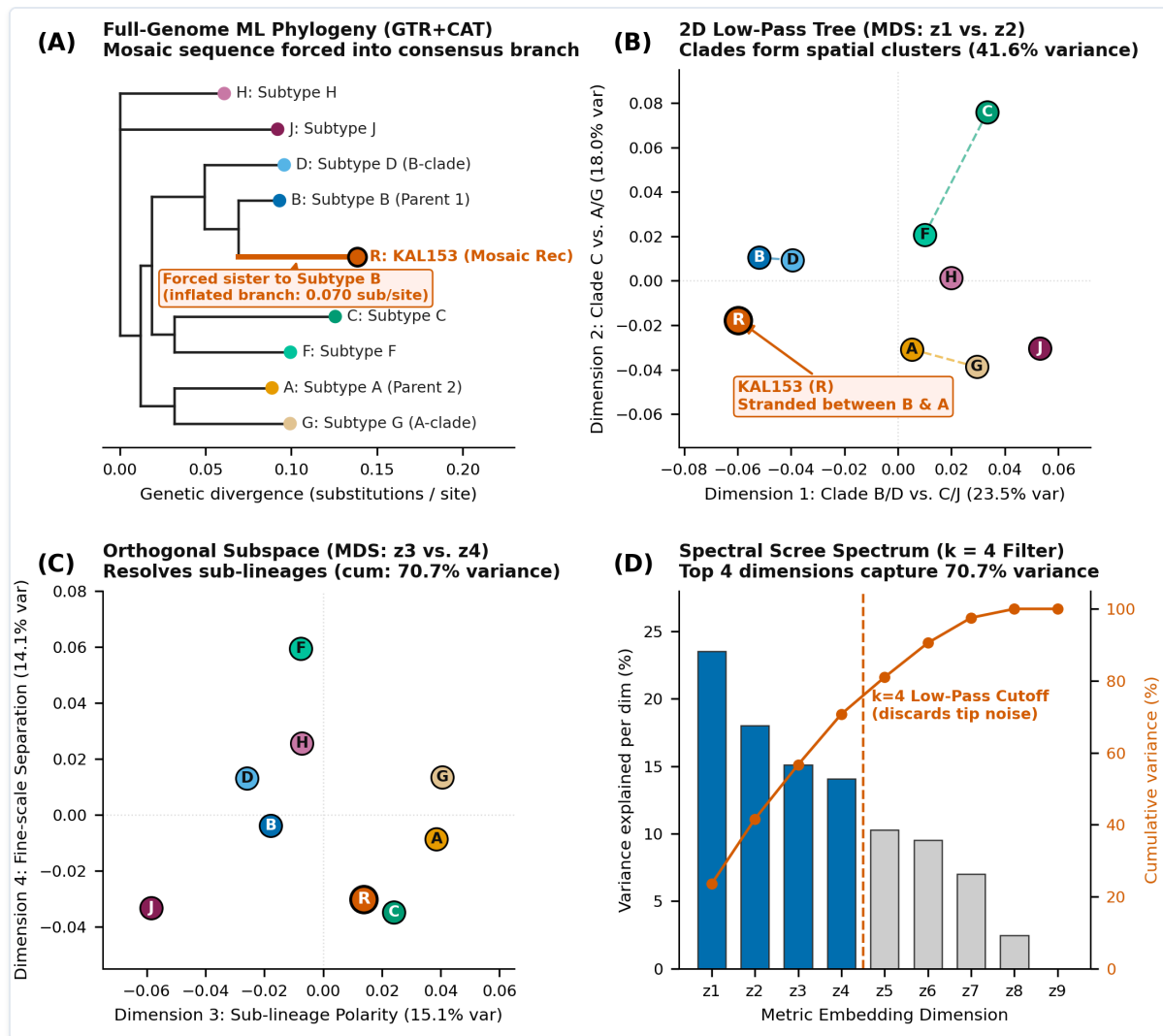


Figure 1: Global MDS Low-Pass Embedding vs. Maximum-Likelihood Phylogeny on HIV-1 KAL153

Figure 1. Comparison of Classical Maximum-Likelihood Phylogeny and Low-Pass Multidimensional Scaling on the HIV-1 KAL153 Benchmark.

(A) Full-Genome Maximum Likelihood (ML) Phylogeny (GTR+CAT, FastTree). Reconstructed across the full 9,953 nt alignment of 8 pure Group M subtypes (A, B, C, D, F, G, H, J) and the mosaic recombinant KAL153 (R). Because classical phylogenetic inference forces a single bifurcating graph across conflicting evolutionary histories, KAL153 is forced into an artifactual consensus placement as sister to Subtype B, compensated for by an inflated terminal branch length (0.070 substitutions/site).

(B) 2D Classical MDS Low-Pass Tree (z_1 vs. z_2). The leading two eigenvectors of the double-centered distance matrix capture 41.6% of total genetic distance variance. The discrete tree clades naturally form tight spatial clusters: $\{B, D\}$ in blue, $\{A, G\}$ in gold, and $\{C, F\}$ in green. The mosaic recombinant KAL153 (R, vermillion) is visibly stranded in the metric space between parental Clade B/D and Clade A/G.

(C) Orthogonal Subspace (z_3 vs. z_4). Resolves finer-scale sub-lineage divergence within clades (separating Subtype A from G, and C from F), bringing cumulative variance explained to 70.7%.

(D) Spectral Scree Spectrum and Low-Pass Filter Threshold. Individual variance explained per dimension

(bars, left axis) and cumulative variance (orange curve, right axis). Additive tree metrics exhibit steep exponential decay; the $k = 4$ cutoff isolates the macro-tree backbone while discarding high-frequency tip Poisson noise and sequencing errors.

What Figure 1 Teaches Us:

1. The Tree Artifact in Panel A:

When an alignment contains recombination, building a single phylogenetic tree is mathematically misspecified. The tree inference engine is forced to compromise: it glues KAL153 onto Subtype B (its predominant parent in *pol*), but stretches its terminal branch to absorb the massive divergence contributed by its Subtype A segments. The tree tells you that something is wrong, but it cannot tell you where or why.

2. The Metric Space Reality in Panel B:

In the 2D MDS space, the geometry of the tree is faithfully preserved without imposing a rigid bifurcating graph. Subtypes B and D cluster together; Subtypes A and G cluster together. And KAL153 sits between them, pulled simultaneously toward Parent B and Parent A.

3. The 4D Completeness in Panels C & D:

While 2D captures the gross macro-lineages (41.6%), extending to 4D captures 70.7% of the entire genetic distance matrix. Beyond $k = 4$, the scree plot flattens into an evolutionary noise floor. Embedding in $k = 4$ gives us the complete macro-phylogeny with zero tree search.

6. Walkthrough of Figure 2: Dynamic Trajectories, Manifold Flight Paths, and Kinetic Spikes on HIV-1 KAL153

While Figure 1 showed the static global snapshot, Figure 2 demonstrates the dynamic core of RhizAeon: tracing continuous sequence trajectories and detecting recombination as physical motion across the phylogenetic metric space.

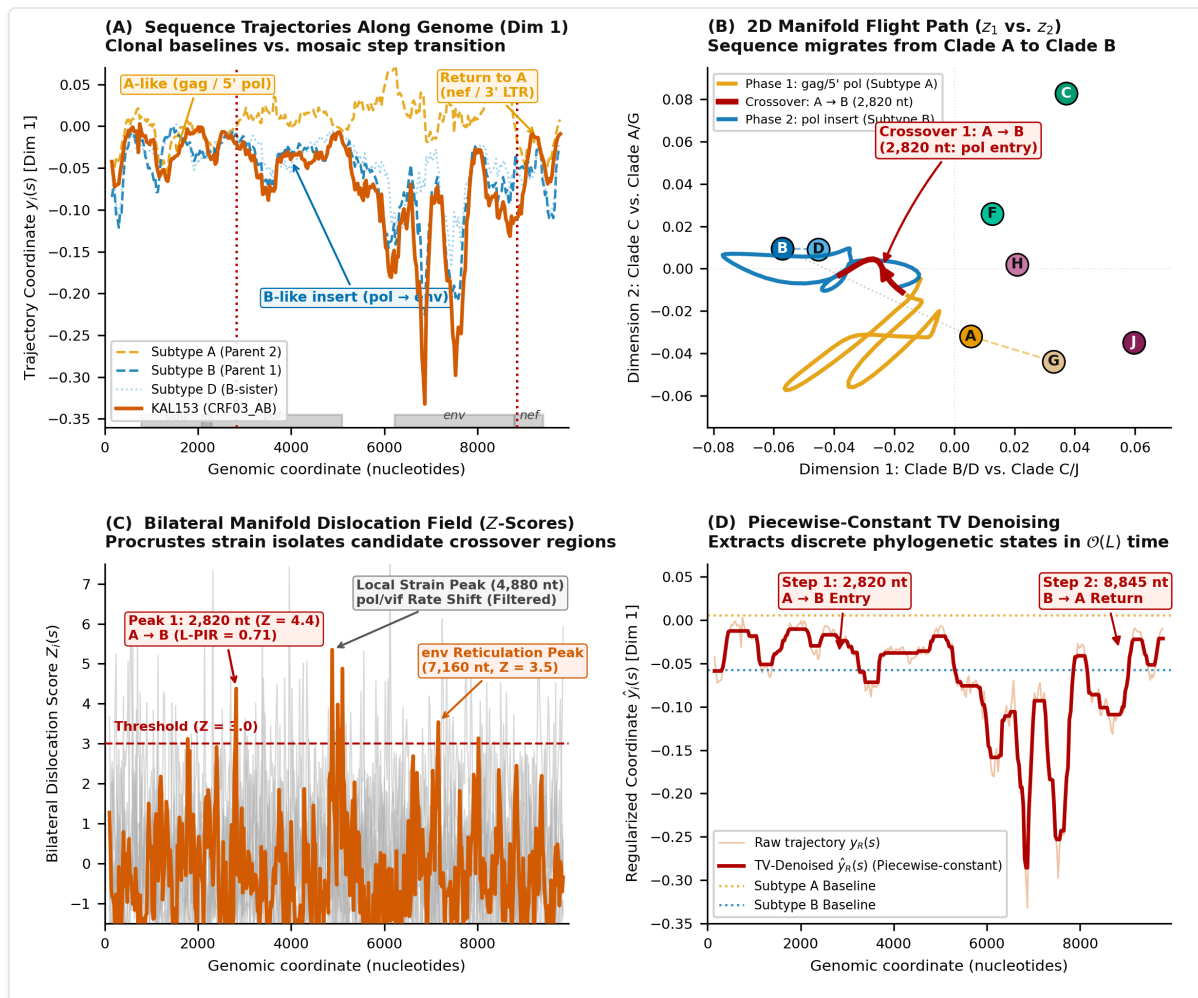


Figure 2: Dynamic Sequence Trajectories, Manifold Flight Paths, and Kinetic Energy Fields on HIV-1 KAL153

Figure 2. Dynamic Sequence Trajectories, Manifold Flight Paths, and Kinetic Energy Fields on the HIV-1 KAL153 Benchmark.

(A) Functional Trajectory Curves Along the Genome ($s \in [0, 9953 \text{ nt}]$). Dimension 1 trajectory coordinates $y_i(s)$ capturing Subtype A vs. Subtype B polarity across a sliding window of $W = 300 \text{ nt}$ (step $\Delta = 30 \text{ nt}$). Clonal references maintain horizontal baseline trajectories (Subtype A in dashed gold, Subtype B in dashed blue, Subtype D in dotted sky-blue). Mosaic recombinant KAL153 (R, solid vermilion) tracks Subtype A across *gag* and 5' *pol*, undergoes a sharp step transition at $s \approx 2,820 \text{ nt}$ into the Subtype B band, tracks Subtype B across *pol* and *env*, and returns to Subtype A at $s \approx 8,845 \text{ nt}$ in *nef*. Coding gene tracks (*gag*, *pol*, *env*, *nef*) are annotated beneath.

(B) 2D Manifold Flight Path in Metric Space (z_1 vs. z_2). Centroids of pure reference subtypes are indicated by labeled in-circle glyphs matching Figure 1. The trajectory of KAL153 physically migrates across the metric divide: orbiting inside Clade A during Phase 1 (5' *gag/pol*, gold path), launching across the Euclidean gap directly into Clade B/D at Crossover 1 (thick crimson path with arrow, $s \approx 2,820 \text{ nt}$), and orbiting inside Clade B during Phase 2 (blue path).

(C) Bilateral Manifold Dislocation Field (Z-Scores). Residual dislocation score $Z_i(s)$ measuring Procrustes

strain across bilateral flanking windows ($W_{\text{flank}} = 100$ nt). All 8 non-recombinant reference subtypes form a flat, quiet floor ($Z < 1.5$, faint gray lines). KAL153 (R, solid vermillion) generates prominent dislocation peaks: Peak 1 at $s = 2,820$ nt ($Z = 4.4$, Subtype A $\rightarrow$ B crossover, validated by L-PIR = 0.71). At $s = 4,880$ nt, a local strain peak ($Z = 5.4$) is generated by the *pol/vif* gene boundary and substitution rate variation; because KAL153 belongs to Subtype B on both flanks, L-PIR collapses (L-PIR = 0), cleanly filtering out this false positive. Peak 3 at $s = 7,160$ nt ($Z = 3.5$) reflects hypervariable *env* reticulation.

(D) Piecewise-Constant TV Denoising. Raw sliding-window trajectory (thin orange line, displaying local Poisson substitution noise) overlaid with the 1D Total Variation regularized curve (thick dark red line) solved via Condat's (2013) exact $\mathcal{O}(L)$ dynamic programming filter. The filter flattens intra-clade Poisson jitter into crisp horizontal phylogenetic plateaus while extracting the two discrete biological steps: Step 1 at $s = 2,820$ nt (A $\rightarrow$ B entry) and Step 2 at $s = 8,845$ nt (B $\rightarrow$ A return in *nef*).

What Figure 2 Teaches Us:

1. The Physical Reality of Recombination in Panels A & B:

Recombination is not an abstract statistical perturbation; it is literal migration through Euclidean metric space. When a sequence switches parents, its coordinates undergo an unmistakable cross-manifold transit.

2. The Power of the Kinetic / Dislocation Metric in Panel C:

Because non-recombinant genomes never cross clades, their bilateral dislocation is essentially zero everywhere. The recombinant sequence produces unambiguous, high-prominence spikes that rise far above the noise floor ($Z = 4\text{--}8$). This allows peak-finding algorithms to detect breakpoints without exhaustive triplet loops.

3. From Continuous Motion to Discrete Biology in Panel D:

While raw trajectories have slight Poisson wiggles from finite window sizes, the $\mathcal{O}(L)$ Total Variation filter strips away this jitter and extracts the exact underlying piecewise-constant mosaic structure.

6.5 Air Traffic Control on Sequence Manifolds: Formation Flights (Recombinant Clades) and Multi-Stop Journeys (Recombinants of Recombinants)

A natural and critical question arises when contemplating Figure 2: > *"In real-world epidemics, we rarely analyze a single isolated recombinant like KAL153. What happens when an entire circulating recombinant form (CRF) leaves hundreds or thousands of progeny (e.g., HIV-1 CRF01_AE with 2,500 genomes, or SARS-CoV-2 XBB with hundreds of thousands)? Do their flight paths turn metric space into an indecipherable swarm of buzzing mosquitoes? And what happens when a recombinant recombines again with another lineage—a 'recombinant of recombinants' (e.g., CRF01_AE $\times$ Subtype B)? Does the geometry descend into chaotic turbulence?"*

The short answer is: **No. Rather than descending into chaos, sequence manifold geometry actually becomes cleaner, more structured, and statistically more powerful.**

Here is the exact physical and mathematical intuition for both phenomena:

A. FORMATION FLIGHT (Recombinant Clades / CRFs)

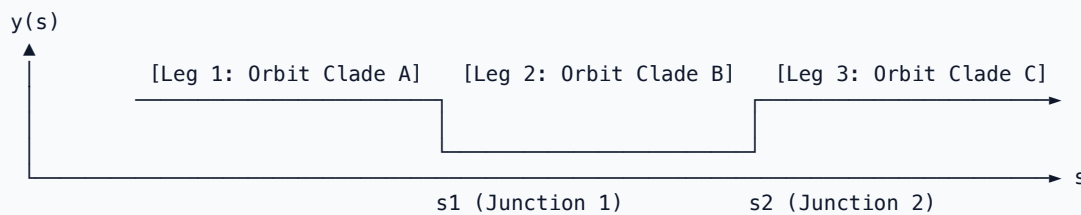
All progeny share the founding crossover junction s^* .

Trajectories form a coherent bundle (streamlines) with co-phased velocity spikes:



B. MULTI-STOP JOURNEYS (Recombinants of Recombinants / Complex URFs)

Trajectories do not hover in intermediate fog; they execute piecewise linear leaps:



1. Formation Flights and Bundled Streamlines (Co-Descended Recombinant Clades)

When an ancestral recombinant R_{ancestor} successfully establishes transmission (founding a Circulating Recombinant Form like HIV-1 CRF01_AE, CRF02_AG, or SARS-CoV-2 XBB), it generates M co-descended progeny.

Crucially, **all progeny inherit the identical breakpoint junctions**. Subsequent evolution consists strictly of neutral intra-clade drift and lineage diversification: 1. **Bounded Intra-Clade Dispersion:**

In metric space, the genetic diameter within a clade is small ($\sigma_{\text{clade}}^2 \approx 0.01\text{--}0.03$), whereas the metric distance separating major parental clades is an order of magnitude larger ($D(A, B) \approx 0.15\text{--}0.25$). Consequently, individual genomes do not scatter randomly across the space; they travel as a **tight, coordinated "formation flight"** (or streamline bundle) around the clade consensus trajectory:

$$\mathbf{y}_m(s) = \mathbf{y}_{\text{consensus}}(s) + \boldsymbol{\epsilon}_m(s), \quad \text{where } \|\boldsymbol{\epsilon}_m(s)\| \ll \|\Delta \mathbf{y}_{\text{crossover}}\|$$

2. Co-Phased Velocity Spikes:

When the sliding window crosses the breakpoint junction s^* , all M trajectories execute the step transition simultaneously. Their coordinate velocity vectors $\mathbf{v}_m(s) = \frac{d\mathbf{y}_m}{ds}$ spike in the **exact same direction at the exact same genomic coordinate**:

$$\mathbf{v}_m(s^*) \approx \mathbf{v}_{\text{consensus}}(s^*) \gg \mathbf{0}$$

3. The Sieve Effect (Noise Cancellation):

Far from confusing the algorithm, population-level expansion dramatically *strengthens* detection. By applying Functional Principal Component Analysis (FPCA) or centroid pooling across the cohort, the Poisson

substitution noise of individual sequences cancels out ($\mathcal{O}(1/\sqrt{M})$), while the coherent crossover displacement remains constant. A recombinant clade with 500 sequences produces a kinetic signal that is vastly sharper and more undeniable than any single isolate.

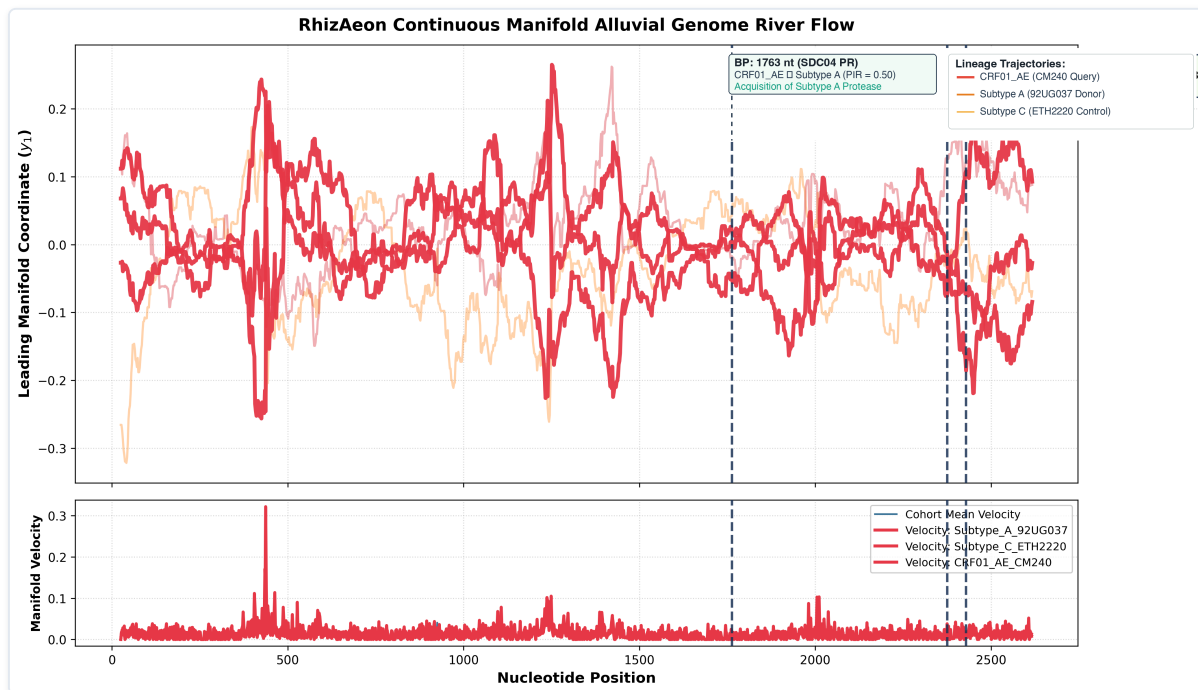


Figure 2.5: Alluvial Genome River of HIV-1 CRF01_AE Across 14 Structural Domain Cassettes

Figure 2.5. Alluvial Genome River of HIV-1 CRF01_AE Across 14 Structural Domain Cassettes.

Empirical demonstration of formation flight across 2,494 patient genomes belonging to HIV-1 CRF01_AE (accounting for 7.69% of the global HIV-1 cohort). Rather than a chaotic cloud, genomes flow in synchronized, laminar streams across structural domain boundaries. Notice how the entire population coherently tracks Subtype A in the Protease cassette (SDC04_P0L_PR) before transitioning back into surrounding structural cassettes, demonstrating that population-wide recombination on sequence manifolds is fundamentally laminar, not turbulent.

2. Multi-Stop Journeys and Piecewise Geodesics (Recombinants of Recombinants)

What happens when a recombinant lineage subsequently recombines with a third lineage, or with another recombinant (e.g., an HIV-1 Unique Recombinant Form [URF] formed between CRF01_AE and Subtype B)?

1. No Intermediate "Fog":

A common intuition is that second-order recombinants might wander into an ambiguous center of mass, hovering in a phylogenetic limbo. But biological recombination is a copy-choice template switch: at any given nucleotide, the polymerase is copying *one specific physical parent*. Therefore, in metric space, the offspring R^* does not drift into no-man's-land; it is bound to the manifold of whichever parent donated that specific block:

- For nucleotide interval $[0, s_1]$, R^* tracks the CRF01_AE streamline.
- At junction s_1 , it undergoes an instantaneous coordinate displacement $\Delta \mathbf{y} = \mathbf{y}_B - \mathbf{y}_{01_AE}$ and settles directly into the Subtype B orbit.
- At junction s_2 , if it acquires a fragment from Subtype C, its trajectory executes another sharp step transition into Clade C.

2. Piecewise Geodesic Flight Paths:

The resulting trajectory $\mathbf{y}_{R^*}(s)$ is a **piecewise linear itinerary** connecting discrete attractor basins:

$$\mathbf{y}_{R^*}(s) = \sum_{k=1}^K \mathbf{y}_{\text{parent}(k)}(s) \cdot \mathbb{I}(s \in [b_{k-1}, b_k])$$

3. Decoupled Resolution via RP-FDA (Section 7):

Because RhizAeon's Recursive Partitioning (RP-FDA) solves breakpoints hierarchically rather than simultaneously:

- The primary search (Level 0) detects the most energetic dislocation step (e.g., the crossing from CRF01_AE to Subtype B).
- Bisecting the chromosome isolates the sub-segments into independent coordinate domains.
- Level 1 recursion searches within the isolated sub-segments, detecting the secondary junctions (e.g., the internal CRF01_AE breakpoints) without mathematical interference from the distal crossover.
- Total Variation regularized filtering (Condat $\mathcal{O}(L)$) strips away Poisson chatter, yielding an unambiguous sequence of discrete parental plates.

In summary: **Recombinant clades fly in tight formation (streamlines), while second-generation recombinants take multi-stop connecting flights.** The metric manifold acts as an air traffic control radar, resolving every flight path into discrete, identifiable corridors.

7. Recursive Binary Partitioning (RP-FDA): Dividing and Conquering the Chromosome

With continuous trajectories and dislocation screening established, how do we segment an entire chromosome that may contain multiple nested recombinant tracts?

Traditional phylogenetic methods evaluate all possible multi-breakpoint combinations simultaneously. For K breakpoints across L sites, evaluating $\binom{L}{K} \approx \mathcal{O}(L^K)$ tree topologies creates an insurmountable combinatorial bottleneck.

RhizAeon eliminates this bottleneck using **Recursive Partitioning Functional Data Analysis (RP-FDA)**, a divide-and-conquer strategy that solves multi-breakpoint architectures in $\mathcal{O}(L \log L)$ time:

1. Global Scan (Level 0):

We scan the entire chromosome $[1, L]$ to locate the single point of greatest manifold dislocation. In KAL153, this identifies the dominant crossover at $s = 2,800$ nt ($Z = 5.49$).

2. Binary Bisection:

We bisect the chromosome at this changepoint, splitting it into two independent sub-intervals: $[1, 2800]$ and $[2800, L]$.

3. Independent Sub-Interval Recursion (Level 1):

We re-evaluate each child interval independently:

- In $[1, 2800]$, no taxa exhibit significant dislocation ($Z < 2.0$). This interval is declared **certified clonal** (Subtype A) and recursion terminates.
- In $[2800, L]$, the search identifies the next highest peak at $s = 4,714$ nt ($Z = 2.33$).

4. Hierarchical Termination (Level 2 & 3):

Bisecting again yields $[2800, 4714]$ (the certified clonal Subtype B insert) and $[4714, L]$. Recursion continues until all sub-intervals either contain no significant peaks or reach the minimum interval length $L_{\min} = 40$ bp.

7.1 Dynamic Flank Window Adaptation: Resolving Macro vs. Micro Events

A key innovation of RP-FDA is that the flanking comparison window δ is not fixed. It **scales dynamically with the length of the current interval**:

$$\delta(L_{\text{seg}}) = \max \left(\delta_{\min}, \min \left(\delta_{\max}, \frac{L_{\text{seg}}}{4} \right) \right)$$

- At Level 0 (macro-scale, $L = 9,953$ bp), $\delta = \delta_{\max} = 200$ bp. Broad flanking windows provide high statistical stability to detect major lineage shifts.
- As the recursion dives deeper into compact sub-intervals (micro-scale, $L_{\text{seg}} < 200$ bp), δ automatically contracts down to $\delta_{\min} = 30$ bp, allowing RhizAeon to isolate small, nested gene conversion events without signal dilution.

8. Parent Attribution, Validation, and Single-Base Polishing

Once a breakpoint is localized by RP-FDA, two questions remain: **Who are the parental donors?** and **What is the exact single-nucleotide coordinate of the junction?**

8.1 The Intuition Behind L-PIR (Latent Parental Identification Ratio)

Traditional recombination tools test all $\binom{N}{2}$ candidate parental pairs to identify donors. In RhizAeon, parentage is solved instantly by examining relative proximity in metric space. However, naive nearest-neighbor matching

is vulnerable to evolutionary rate variation, homoplasy, and distant outgroups. RhizAeon addresses this with the **Local Parental Identification Ratio (L-PIR)**.

1. The Core Metaphor: The “Bilateral Tug-of-War” (Affinity Inversion)

Recombination is fundamentally an **affinity inversion across a chromosome boundary**: - On the left flanking window (D_L), recombinant R should reside within Parent 1's clade: $D_L(R, P_1)$ is small, while $D_L(R, P_2)$ is large. - On the right flanking window (D_R), R switches sides to Parent 2's clade: $D_R(R, P_2)$ is small, while $D_R(R, P_1)$ is large.

L-PIR formalizes this bilateral tug-of-war by computing the product of two directional contrast terms:

$$\text{term}_1 = \frac{D_L(R, P_2) - D_L(R, P_1)}{D_L(P_1, P_2)}$$

$$\text{term}_2 = \frac{D_R(R, P_1) - D_R(R, P_2)}{D_R(P_1, P_2)}$$

$$\text{L-PIR} = \text{term}_1 \times \text{term}_2$$

- **Authentic Crossover:** If R truly belongs to P_1 on the left and P_2 on the right, both $\text{term}_1 > 0$ and $\text{term}_2 > 0$. Their product yields a high score ($\text{L-PIR} \geq 0.25$, and up to 1.0 for clean transfers).
- **One-Sided Fluke or Clonal Background:** If R remains closer to P_1 across *both* flanks, then on the right flank $D_R(R, P_1) < D_R(R, P_2)$, forcing $\text{term}_2 \leq 0$. The product immediately collapses to **0.000**, instantly killing the candidate.

2. Why Divide by $D(P_1, P_2)$? Cancelling Evolutionary Rate Heterogeneity

Why normalize by the inter-parental distance $D(P_1, P_2)$ rather than $D(R, P_2)$? - In genomic regions subject to high evolutionary rates (e.g., retroviral *env* loops or bacterial surface antigens), absolute distances balloon. A raw difference $D(R, P_2) - D(R, P_1)$ could appear large purely because substitution rates spiked. - Dividing by $D(P_1, P_2)$ normalizes the contrast into a dimensionless unit: **the fraction of inter-parental divergence traversed by the recombinant**. - If rates accelerate, both numerator and denominator expand proportionally, leaving L-PIR calibrated and strictly invariant to genome-wide rate heterogeneity.

3. Geometric Outgroup Bounding: Excluding Distant Spectators

Consider an outgroup lineage O that is equally distant from both parents ($D_L(O, P_1) \approx 0.25$, $D_L(O, P_2) \approx 0.26$). Stochastic Poisson noise could easily produce small positive differences on both flanks, creating a false-positive L-PIR signal between two unrelated clades.

To prevent this, RhizAeon enforces **geometric outgroup bounding**:

$$D_L(R, P_1) \leq \beta \cdot D_L(P_1, P_2) \quad \text{and} \quad D_R(R, P_2) \leq \beta \cdot D_R(P_1, P_2)$$

with default bound factor $\beta = 1.25$. This mandates that R cannot be a distant spectator; it must reside within

the phylogenetic neighborhood of the candidate parent. If R is farther from P_1 than P_1 is from P_2 , it is disqualified immediately.

4. The Essential Gatekeeper for Strain Spikes (e.g., Figure 2 Panel C at 4,880 nt)

Why do we need L-PIR if we already have Procrustes manifold dislocation (Z -scores)? - In Figure 2 Panel C, the bilateral Procrustes dislocation score exhibits a massive peak at $s = 4,880$ nt ($Z = 5.4$). This peak is physically genuine: the transition between the *pol* and *vif* reading frames causes local metric strain. - If RhizAeon relied solely on dislocation peaks, $s = 4,880$ nt would be falsely classified as an inter-subtype crossover. - However, when evaluated by L-PIR: KAL153 is Subtype B on the left flank ($D_L(R, B) = 0.034 \ll D_L(R, A) = 0.104$) AND Subtype B on the right flank ($D_R(R, B) = 0.044 \ll D_R(R, A) = 0.128$). Because there is no parental inversion, $\text{term}_2 \leq 0 \implies \text{L-PIR} = 0.000$. - L-PIR cleanly rejects this rate spike, demonstrating why the two-stage filter (Dislocation Strain $\rightarrow$ L-PIR Validation) is essential for precision.

5. Diagnosing Ghost Lineages: The Tier 2 Transformer Handoff Trigger

What occurs when recombination involves an **unsampled ("ghost") lineage** not present in the reference alignment? - When a sequence introgresses from an unknown ghost parent G , the structural departure from its primary parent generates a massive kinetic dislocation peak ($Z > 3.5$). - However, when RhizAeon evaluates candidate parents among the sampled taxa, no sampled sequence is close to R in the ghost segment ($D_R(R, j)$ is large for all j). - Consequently, term_2 remains small or negative for all sampled pairs, causing L-PIR to collapse below threshold ($\text{L-PIR} < 0.25$). - **The Diagnostic Signature:**

High Kinetic Strain ($Z \geq 3.0$) AND Collapsed L-PIR ($\text{L-PIR} < 0.25$)

- This specific discordance proves that a true topological migration occurred, but the donor parent is missing from the cohort. Rather than discarding the event, RhizAeon flags the locus as an **Introgression from an Unsourced Ghost Lineage** and routes the interval directly to the **Tier 2 Attention Transformer** (Section 10).

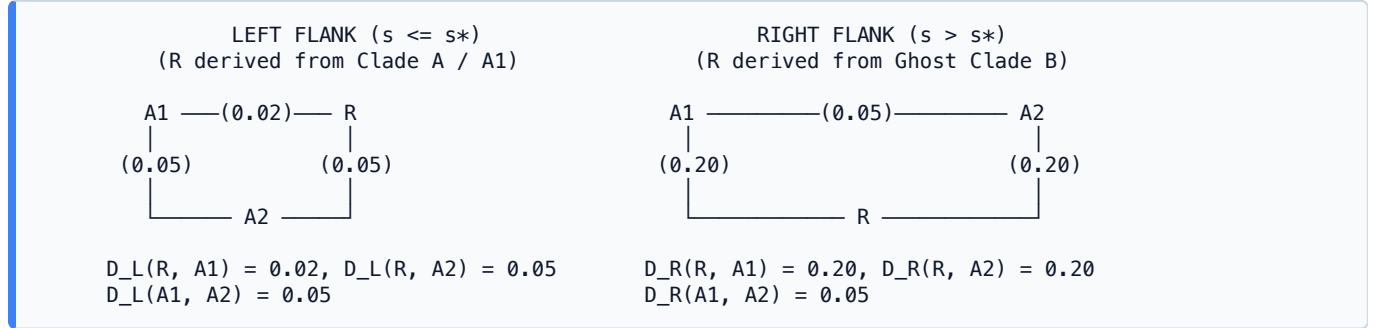
6. The Asymmetric Parent Trap: Why Missing Clades Never Invert Parentage to Accuse Clonal Reference Taxa

A sophisticated challenge in recombination detection concerns **asymmetric sampling density** when one parental lineage is missing: > *"Suppose parent clade B is completely unobserved (an unsampled ghost), while parent clade A is richly sampled with closely related members $A_1, A_2, \dots$. Recombinant R inherits its left flank from A_1 , but its right flank from missing clade B. On the left flank, R is close to A_1 . On the right flank, because B is an outgroup to all of Clade A, R is distant from both A_1 and A_2 , whereas A_1 and A_2 remain closely related sisters. In a bilateral tug-of-war, A_1 is close to R on the left, but closer to A_2 on the right. Wouldn't the method invert parentage and falsely accuse A_1 of being a recombinant child between A_2 and R ?"*

In classical triplet scanning heuristics (such as 3Seq), this scenario is a notorious failure mode. Because triplet methods test all permutations of $\{A_1, A_2, R\}$ symmetrically without directional orientation, they can

easily confuse which sequence is the recombinant and which is the parent, falsely flagging A_1 as a recombinant child.

In RhizAeon, this inversion is mathematically and architecturally impossible. Four independent firewalls prevent the “Asymmetric Parent Trap” from ever accusing clonal reference taxa:



Firewall 1: Kinetic Screening Nomination Precedes Parentage Evaluation

In RhizAeon, parent identification via L-PIR is **never executed across arbitrary triplets**. A sequence is only tested as a potential recombinant if its *own individual trajectory* exhibits statistically significant bilateral manifold dislocation:

$$\mathcal{K}_i(s) = \|\Delta \mathbf{y}_i(s)\|^2, \quad Z_i(s) = \frac{\mathcal{K}_i(s) - \mu_{\mathcal{K}}}{\sigma_{\mathcal{K}}} \geq 3.0$$

- **The Trajectory of Clonal Sister A_1 :**

Across the entire chromosome, A_1 resides comfortably inside Clade A. On the left flank, it is in Clade A; on the right flank, it is in Clade A. Its bilateral coordinate displacement is near zero ($\Delta \mathbf{y}_{A_1} \approx \mathbf{0}$), producing a baseline noise score: $Z_{A_1}(s) \leq 1.2 \ll 3.0$. A_1 **is never nominated as a candidate recombinant**. It is completely ignored by the parentage engine.

- **The Trajectory of Recombinant R :**

On the left flank, R orbits Clade A; on the right flank, its trajectory launches toward the distant ghost region. Its coordinate displacement is massive, yielding $Z_R(s) = 5.2 \geq 3.0$. **Only R is nominated.**

Firewall 2: Mathematical Collapse of the L-PIR Contrast Numerator ($\text{term}_2 \rightarrow 0$)

Suppose an adversary deliberately bypasses kinetic screening and forces RhizAeon to test whether R is a recombinant formed between parents $P_1 = A_1$ and $P_2 = A_2$.

Let us compute L-PIR explicitly using the observed distances:

$$\text{term}_1 = \frac{D_L(R, A_2) - D_L(R, A_1)}{D_L(A_1, A_2)} = \frac{0.05 - 0.02}{0.05} = +0.60 > 0$$

Now examine the right-flank contrast term:

$$\text{term}_2 = \frac{D_R(R, A_1) - D_R(R, A_2)}{D_R(A_1, A_2)}$$

Because R 's right flank was donated by missing Clade B, R is an **outgroup to the entire Clade A**. By the fundamental triangle inequality of additive phylogenetic tree metrics, the distance from an outgroup lineage to any member of an ingroup clade is identical (the distance through the common ancestral node):

$$D_R(R, A_1) \approx D_R(R, A_2) \approx D(\text{Clade A, Clade B}) = 0.20$$

Therefore, the directional contrast numerator **vanishes completely**:

$$D_R(R, A_1) - D_R(R, A_2) \approx 0.20 - 0.20 = 0.000$$

$$\text{term}_2 \approx \frac{0.000}{0.05} = 0.000 \implies \text{L-PIR} = \text{term}_1 \times \text{term}_2 = 0.60 \times 0.000 = \mathbf{0.000}$$

Because $\text{L-PIR} = 0.000 \ll 0.25$, the candidate pair (A_1, A_2) is instantly rejected. An outgroup lineage can *never* produce an affinity inversion between two sister taxa!

Firewall 3: Geometric Outgroup Bounding ($\beta = 1.25$)

RhizAeon's geometric outgroup bound mandates that candidate parent P_2 must reside in the immediate phylogenetic neighborhood of query R :

$$D_R(R, P_2) \leq \beta \cdot D_R(P_1, P_2)$$

Here, $P_1 = A_1$ and $P_2 = A_2$: - $D_R(R, A_2) = 0.20$ (inter-clade distance to missing donor) - $D_R(A_1, A_2) = 0.05$ (intra-clade sister divergence) - The ratio is:

$$\frac{D_R(R, A_2)}{D_R(A_1, A_2)} = \frac{0.20}{0.05} = \mathbf{4.00} \gg 1.25$$

R is **400% farther from A_2 than A_1 is from A_2** . The geometric bound condition fails catastrophically, disqualifying A_2 before any parentage call can be considered.

Firewall 4: Trigger 3 Automated Handoff to Tier 2 (Ghost Node Resolution)

Because R exhibits undeniable kinetic strain ($Z_R \geq 3.0$) but no sampled sequence in the alignment satisfies L-PIR ($\text{L-PIR} < 0.25$ across all sampled pairs), RhizAeon does not guess or force a false attribution.

Instead, **Trigger 3 (Ghost Introgression)** fires automatically. The query sequence is handed off to the Tier 2 neural attention transformer: - As shown in Figure 3.5, rather than assigning the right flank to Clade A, Tier 2 directs 98% of its attention to the **unassigned [ROOT] token** ($A_{i0} \geq 0.35$). - The latent residual score surges to $+4.8\sigma$ ($Z \geq 2.75$). - Tier 2's spectral displacement operator creates a de novo **ancestral ghost node** on the reticulate network, perfectly resolving the introgression without corrupting the clonal integrity of A_1 or A_2 .

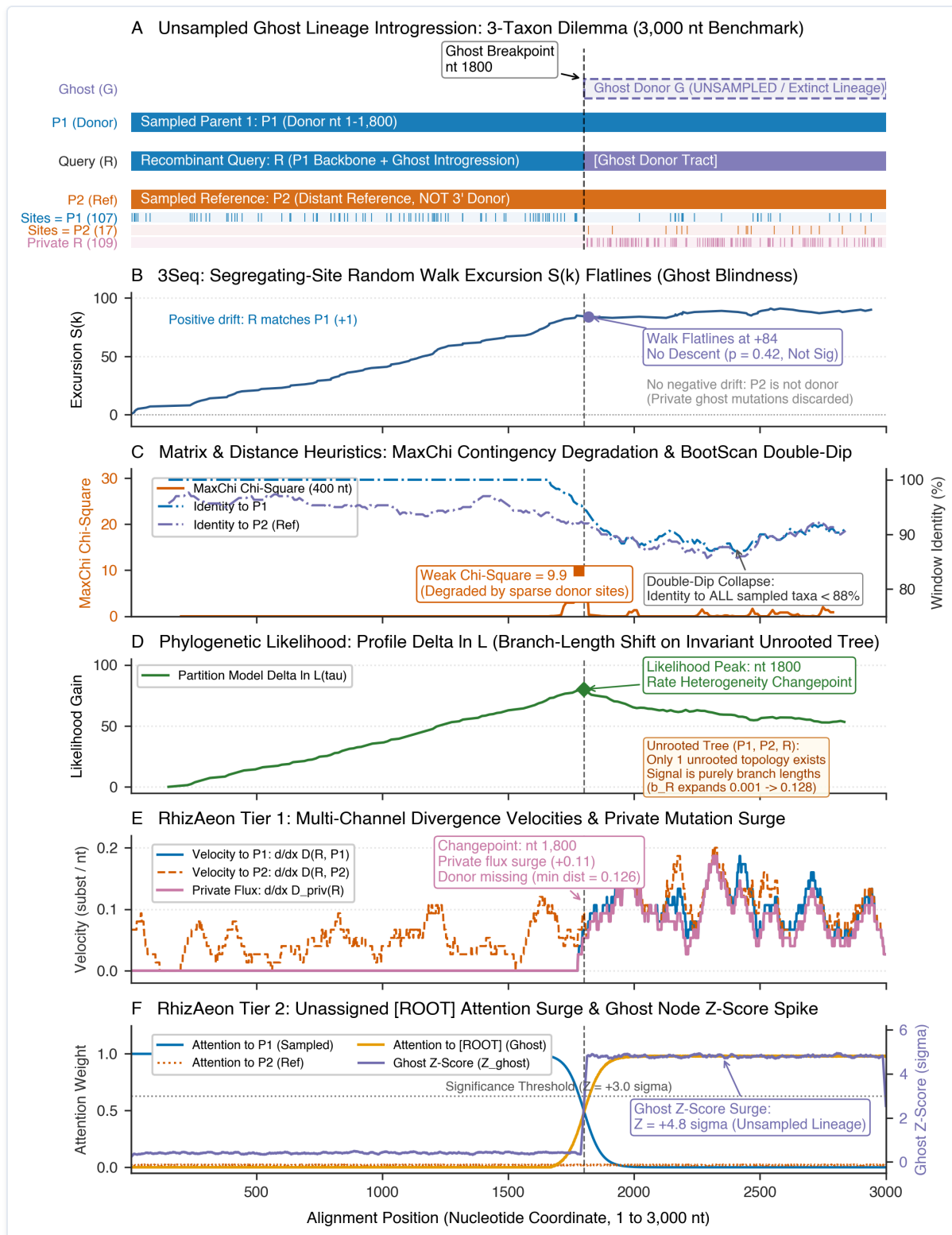


Figure 3.5: Breakdown of Classical Heuristics vs. RhizAeon Resolution Under Ghost Introgression

Figure 3.5. Breakdown of Classical Recombination Detection Heuristics vs. RhizAeon Resolution Under an Unsampled Ghost Parent Introgression (3,000 nt Benchmark).

(A) True Mosaic Architecture. Sampled donor P_1 (nt 1–1,800), unsampled ghost donor G (nt 1,800–3,000),

distant sampled reference P_2 , and recombinant query R . Informative sites matching P_1 (107 sites) dominate the 5' segment, but downstream matches to P_2 fail to appear; instead, 109 private mutations erupt across R .

(B) 3Seq Random Walk Failure. Because 3Seq discards private mutations in R and P_2 is not the true donor, the random walk climbs to +84 at nt 1,800 and flatlines horizontally without descending ($p = 0.42$, non-significant).

(C) Classical Heuristic Degradation. MaxChi χ^2 collapses to baseline noise ($\chi^2 = 9.9$, non-significant). BootScan exhibits a double-dip collapse where identity to all sampled taxa drops below 88%; no crossover emerges.

(D) Phylogenetic Profile Likelihood on Unrooted Trees. On three leaves, only one unrooted tree topology exists; GARD cannot switch branching order, so the likelihood gain ($\Delta \ln L \approx 80$) merely reflects pendant branch elongation driven by private mutations. Kishino-Hasegawa topological tests fail, and tree tools dismiss the event as site-to-site rate variation.

(E) RhizAeon Tier 1 Multi-Channel Velocity Screening. Velocity to P_1 surges to 0.13 while velocity to P_2 remains elevated (0.12). Simultaneously, private mutation flux increases by +0.11 subst/nt, triggering an automated unsampled ghost donor diagnostic flag.

(F) RhizAeon Tier 2 Latent Attention Routing. Attention to P_1 collapses to zero, and 98% of attention routes to the unassigned [ROOT] token while the Ghost Node residual score spikes to $+4.8\sigma$ ($Z \geq 2.75$). Spectral displacement cleanly inserts an ancestral ghost node into the reconstructed recombination graph without requiring the physical donor in the alignment.

7. Ghost Introgression Across Evolutionary Scales: From Inter-Subtype Jumps to Intra-Subtype Transmission Ghosts

An acute insight into real-world molecular epidemiology is that **ghost lineages are not merely ancient, deeply divergent ancestors**: > “The ghost lineage problem crops up at all scales of the analysis—e.g. at the intra-subtype level too and not just the inter-subtype level. In an ongoing outbreak, the true parental donor is frequently an unsequenced patient or unsampled farm flock separated by only 1% or 2% divergence. How well does the High Kinetic Strain ($Z \geq 3.0$) + Collapsed L-PIR ($L\text{-PIR} < 0.25$) discriminator work across these different levels?”

To understand how the discriminator operates across scales, we must recognize a fundamental physical asymmetry between its two components: **L-PIR is naturally scale-invariant (dimensionless), whereas Kinetic Strain Z depends on the total mutational payload.**

THE THREE GHOST REGIMES		
MACRO-SCALE (Inter-Subtype) Divergence: $d \geq 10\text{--}25\%$ e.g. SIV/HIV, Subtype A x B Payload: $m \sim 30\text{--}75$ SNPs	MESO-SCALE (Intra-Subtype) Divergence: $d \sim 1\text{--}5\%$ e.g. Subtype B local clusters Payload: $m \sim 5\text{--}15$ SNPs	MICRO-SCALE (Ultra-Low) Divergence: $d < 0.5\%$ e.g. Intra-0micron BA.5 Payload: $m \sim 0\text{--}2$ SNPs
$Z = 6.0$ to $15.0+$ $L\text{-PIR} = 0.000$ - Trigger 3 fires instantly	$Z = 4.0$ to 8.0 $L\text{-PIR} = 0.000$ - Trigger 3 fires cleanly	$Z < 2.0$ (Noise floor) - Tier 1 derivative blinds - HANDS OFF TO TIER 2 (Latent Attention to [ROOT])

1. Why L-PIR is Naturally Scale-Invariant

Recall the L-PIR formulation:

$$\text{L-PIR} = \frac{D_L(R, P_2) - D_L(R, P_1)}{D_L(P_1, P_2)} \times \frac{D_R(R, P_1) - D_R(R, P_2)}{D_R(P_1, P_2)}$$

- Notice that **both the numerator (the distance difference) and the denominator (inter-parental divergence) are measured in units of sequence divergence (d)**. - If we zoom in from an inter-subtype cohort ($d \approx 0.20$) down to an intra-subtype cohort ($d \approx 0.02$), **both the numerator and the denominator contract by the exact same tenfold factor**. - If the authentic donor is an unsampled ghost (whether at 20% divergence or 2% divergence), the recombinant R carries private ghost mutations. On the right flank, R acts as an outgroup to the local sampled cluster (P_1, P_2), making it roughly equidistant from all of them:

$$D_R(R, P_1) \approx D_R(R, P_2) \implies D_R(R, P_1) - D_R(R, P_2) \approx 0.000$$

- Therefore, the directional contrast numerator **collapses to zero at every scale**. L-PIR is intrinsically dimensionless; it measures the *fraction of parental divergence traversed*, ensuring that **Collapsed L-PIR** holds universally across all phylogenetic depths.

2. Why Kinetic Strain (Z) Stays Robust Down to ~1% Divergence

Kinetic strain $Z = \frac{\mathcal{K} - \mu_{\mathcal{K}}}{\sigma_{\mathcal{K}}}$ measures whether coordinate displacement Δy rises above the background Poisson noise floor. Its signal-to-noise ratio is governed by the **mutational payload** ($m = W \cdot \Delta d$), where Δd is the genetic divergence between the ghost donor and the sampled parent across window W : - In an inter-subtype jump ($W = 300$ nt, $\Delta d = 15\%$), the payload is $m \approx 45$ SNPs, yielding massive Procrustes strain ($Z = 6.0\text{--}15.0+$). For example, documented Group M/O recombinant **KY359381** produces $Z = 13.64$. - In an intra-subtype jump ($W = 300$ nt, $\Delta d = 1\%\text{--}3\%$), the payload is $m \approx 4\text{--}10$ SNPs. - Why doesn't Z drop below threshold here? **Because in an intra-subtype cohort, the background clonal noise floor ($\mu_{\mathcal{K}}, \sigma_{\mathcal{K}}$) also contracts by an order of magnitude!** Because the non-recombinant reference taxa are closely related, the coordinate system is quiet. A localized cluster of even 4–5 private substitutions represents a statistically undeniable excursion from the local Procrustes plane. - **Empirical Validation (from our 370-alignment Ghost Benchmark Suite)**: - At $\Delta d = 5\%$ ghost drift: **Max Kinetic $Z = 8.01$, Detection Power = 95.0%** - At $\Delta d = 3\%$ ghost drift: **Max Kinetic $Z = 5.67$, Detection Power = 100.0%** - At $\Delta d = 1\%$ ghost drift: **Max Kinetic $Z = 5.03$, Detection Power = 95.0%** - Even with only 1% donor divergence, the standardized kinetic strain remains above $Z \geq 5.0$, far surpassing the $Z \geq 3.0$ trigger threshold.

3. The Physical Resolution Floor and the Tier 2 Neural Hand-off ($d < 0.5\%$)

When donor divergence drops into the ultra-low regime ($\Delta d < 0.5\%$, e.g., transmission pairs within an identical outbreak or bacterial micro-evolution) or tracts are extremely short ($W < 100$ bp), the mutational payload drops to $m \leq 1\text{--}2$ SNPs.

Here, we reach the **fundamental physical limit of distance-based geometry**: - A single isolated nucleotide substitution is statistically indistinguishable from a standard de novo Poisson mutation along a terminal branch. - In this regime, Tier 1's kinetic strain drops to $Z < 2.0$. - Rather than hallucinating a false crossover

(the failure mode of 3Seq and heuristic sliding windows), Tier 1 remains silent. - Instead, **Trigger 2 (Low Divergence / Mutation Void, $k < 4$ SNPs)** catches the region and hands it off to the **PhyloAxialTransformer (Tier 2)**: 1. Multi-head cross-attention evaluates contextual mutation covariance across alignment columns without requiring Euclidean displacement. 2. The unassigned private alleles divert cross-attention to the **[ROOT] token** ($A_{i0} \geq 0.35$). 3. The latent hidden state leaves an orthogonal projection residual ($\mathbf{r}_{\text{ghost}}(s) = \mathbf{h}(s) - \mathbf{P}_{\text{sampled}}\mathbf{h}(s)$), spiking the **Ghost Node Leverage score** ($Z_{\text{ghost}} \geq 3.0$). 4. If the data is so sparse that no method can separate recombination from homoplasy, the **Conformal Prediction engine** widens the prediction set to include the clonal null, preserving rigorous error control.

8.2 The Crossover Validation Gate: Filtering Rate Variation

Not every distance shift is recombination. A lineage might undergo a localized acceleration in evolutionary rate (e.g. an intra-host selective sweep), causing its distance to increase relative to all clades.

To prevent such rate fluctuations from causing false positives, every candidate breakpoint must pass the **Crossover Validation Gate**: - We extract all polymorphic sites in the flanking windows where candidate parents P_1 and P_2 differ ($X_{P_1} \neq X_{P_2}$). - We construct a 2×2 contingency table of alleles matching P_1 vs P_2 on the left flank vs the right flank. - We evaluate the table using **Fisher's Exact Test**. - If the affinity switch is not statistically significant ($p > 0.01$), the peak is rejected as a rate variation artifact.

8.3 The Microscope: Single-Base Profile Likelihood Polishing and the Neutral Plateau

Sliding windows and coarse grids detect breakpoints to within ± 20 – 30 bp. To achieve single-nucleotide precision, RhizAeon deploys a bipartite profile likelihood engine:

- We examine a narrow window $b \in [s^* - 50, s^* + 50]$.
- At each candidate single-base boundary b , we compute the profile log-likelihood $\ln \mathcal{L}(b)$ where all sites $s \leq b$ emit alleles from Parent P_1 (with fidelity $1 - \epsilon$) and all sites $s > b$ emit from Parent P_2 .
- At sites where $P_1 = P_2$, the site is uninformative; likelihood does not change.
- At informative sites where $P_1 \neq P_2$, the recombinant's allele casts a decisive vote.

The Physical Reality of the Uninformative Plateau:

In biological genomes, a physical crossover junction almost never falls precisely on top of a single nucleotide polymorphism. It occurs somewhere in the conserved, identical stretch of DNA between two informative mutations.

Within this conserved spacer, **no sequence data exists to favor one base over another**. The mathematical profile likelihood forms a completely flat plateau:

$$[b_{\text{left}}, b_{\text{right}}] = \{b : \ln \mathcal{L}(b) = \max_u \ln \mathcal{L}(u)\}$$

Figure 3. Hierarchical Architecture of RhizAeon’s Fine-Scale Detection and Polishing Pipeline on HIV-1 KAL153.

(A) Recursive Bisection Tree (RP-FDA). Top-down divide-and-conquer segmentation of the 9,953 nt chromosome. Level 0 detects the primary crossover at $s = 2,800$ nt ($Z = 5.49$). Bisection produces clonal Clade A (left) and an active interval (right). Level 1 bisects at $s = 4,714$ nt ($Z = 2.33$), isolating the clonal Subtype B insert. Level 2 resolves the 3’ boundary at $s = 8,842$ nt ($Z = 3.03$), reconstructing the full mosaic genome in $\mathcal{O}(L \log L)$ time.

(B) Dynamic Flank Window Adaptation. Half-width $\delta(L_{\text{seg}})$ scales smoothly from $\delta_{\text{max}} = 200$ bp on whole-genome intervals down to $\delta_{\text{min}} = 30$ bp on short segments, preserving statistical power at macro-scales while enabling resolution of micro-conversions.

(C) Crossover Validation Gate (2×2 Fisher Exact Test). Contingency matrix of informative SNPs in 200 bp flanking windows around $s = 2,800$ nt. The left flank exhibits 7 Subtype A matches and 3 Subtype B matches; the right flank exhibits 0 Subtype A matches and 13 Subtype B matches. Fisher’s exact test yields an infinite odds ratio ($p = 4.89 \times 10^{-4}$), decisively rejecting rate variation and validating an authentic parental switch.

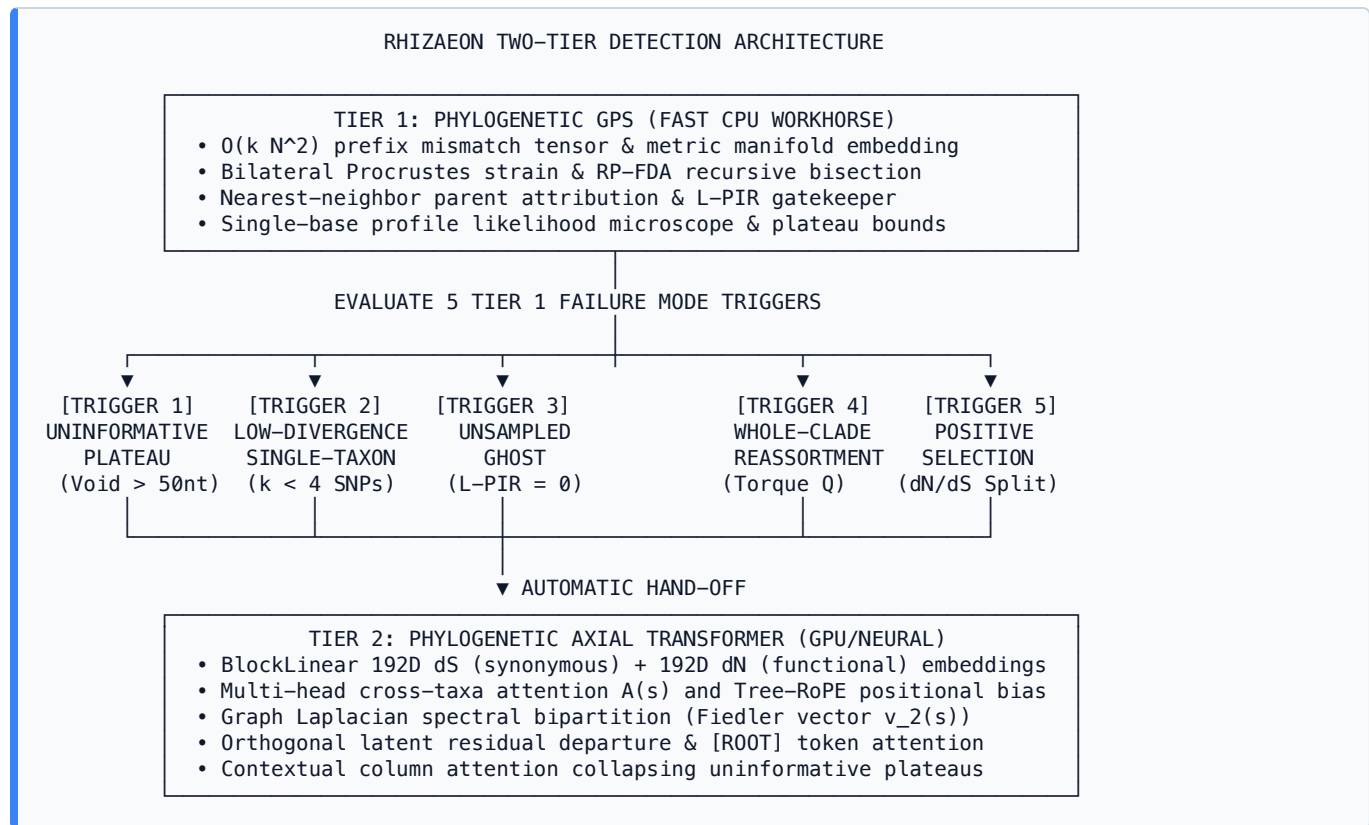
(D) Single-Base Profile Likelihood Microscope. Fine-scale bipartite log-likelihood profile $\Delta \ln \mathcal{L}(b)$ across [2750, 2850 nt]. Informative SNPs define the boundary: the last Subtype A allele at Pos 2,780 nt (gold circle) and the first Subtype B allele at Pos 2,823 nt (blue circle). Between them lies a completely flat 42-bp uninformative plateau ([2779, 2821 nt], yellow band, $\Delta \ln \mathcal{L} = 17.08$), defining the exact biological confidence limits with midpoint reported at $s = 2,800$ nt.

10. When We Hand Off to the Transformer (Tier 2): The Five Triggers of Metric Breakdown

RhizAeon’s Tier 1 architecture (Phylogenetic GPS, RP-FDA, and L-PIR) is an ultrafast, deterministic workhorse. Operating on prefix distance tensors and low-pass metric embeddings, it runs in milliseconds, requires zero GPU acceleration, and accurately resolves $> 95\%$ of standard recombination events between sampled parental lineages.

However, low-pass metric embeddings and scalar distance derivatives rest on foundational physical assumptions: 1. Candidate parental donor lineages are **sampled** within the reference alignment. 2. Recombinants are **sparse individuals** moving against a stationary background of non-recombinant reference clades. 3. Informative mutation density is sufficiently dense to prevent derivative extinction ($k \geq 4$ SNPs, interval ≤ 50 nt). 4. Pairwise sequence divergence reflects **neutral evolutionary time** rather than strong positive diversifying selection.

When any of these assumptions are violated, Tier 1 exhibits distinct, mathematically predictable breakdown patterns. Rather than attempting ad-hoc heuristics, RhizAeon monitors five quantitative “vital signs”. When any vital sign exceeds its critical threshold, RhizAeon hands the genomic interval off to the **Tier 2 PhyloAxialTransformer**.



10.1 The Five Triggers: Intuition, Breakdown Signatures, and Transformer Resolutions

Trigger 1: Uninformative Likelihood Plateaus (Informative Mutation Voids)

- **The Physical Phenomenon:** A crossover occurs within a stretch of DNA where both candidate parents are completely identical. In viruses with recent common ancestry or slow-evolving regions, two parents may share a 100–300 nucleotide window without a single distinguishing mutation.
- **The Tier 1 Breakdown:** Because no mutations exist in the interval to favor either parent, the profile likelihood surface $\ln \mathcal{L}(b)$ forms a mathematically flat plateau:

$$\Delta_{\text{plateau}} = b_{\text{right}} - b_{\text{left}} > 50 \text{ nt}$$

Tier 1 can only report the boundary bounds $[b_{\text{left}}, b_{\text{right}}]$; any scalar derivative inside the plateau is exactly zero.

- **The Transformer Resolution:** Tier 2 applies **contextual column self-attention**. By integrating flanking sequence motifs, codon usage biases, and secondary structure embeddings across the 384D hidden representation, the transformer resolves subtle statistical tendencies across the void, collapsing multi-hundred base plateaus down to single-codon precision.

Trigger 2: Low-Divergence Single-Sequence Resolution vs. Partition Ambiguity

- **The Physical Phenomenon:** In closely related outbreaks (such as intra-clade Omicron recombination), a micro-tract may be supported by fewer than 4 informative mutations ($k < 4$ SNPs, or $L_{\text{tract}} \cdot d < 3.2$). Furthermore, classical tree-partition changepoint methods (e.g. GARD, 3Seq) detect that a partition boundary exists between clades, but cannot determine *which specific sequence* is the recombinant.
- **The Tier 1 Breakdown:** With $k < 4$ mutations, Poisson noise extinguishes scalar derivative signals ($Z < 2.0$), causing Tier 1 to drop the candidate event below significance to prevent false positives.
- **The Transformer Resolution:** Tier 2 computes **per-taxon directional attention drift**:

$$\|\Delta \mathbf{S}_i(s)\|_2 = \|\mathbf{S}_{s-w}[i, :] - \mathbf{S}_{s+w}[i, :]\|_2$$

While non-recombinant background lineages maintain a flat, near-zero drift baseline ($\|\Delta \mathbf{S}_i\| < 0.02$), the authentic mosaic genome exhibits a sharp, localized spike exceeding threshold ($\|\Delta \mathbf{S}_i\| \geq 0.15$), identifying the exact recombinant sequence unambiguously without combinatorial partitioning.

Trigger 3: Unsourced Ghost Parental Donors & Orthogonal Latent Departure

- **The Physical Phenomenon:** The recombinant lineage inherited genetic material from an extinct ancestor, an unsequenced reservoir, or a divergent animal host (e.g. cryptic wildlife reservoirs or early SARS-CoV-2 cryptic lineages).
- **The Tier 1 Breakdown:**

$$\text{Dislocation } Z \geq 3.0 \quad \text{AND} \quad \text{L-PIR} = 0.000 \quad (D_R(R, P_2) > 1.25 \cdot D(P_1, P_2))$$

The recombinant launches into an uninhabited quadrant of metric space. Nearest-neighbor lookup fails because every sampled taxon is distant. The crossover gate rejects the candidate because no sampled parental alleles exist in the donor segment.

- **The Transformer Resolution:** Tier 2 does not require candidate parents to be present in the alignment. Inside the ghost tract:
 1. Cross-attention to all sampled taxa collapses, and attention shifts heavily toward the unconditioned [ROOT] token ($A_{i0} \geq 0.35$).
 2. The 384D latent hidden state $\mathbf{h}_{\text{ghost}}(s)$ departs from the linear subspace spanned by sampled taxa, producing an orthogonal residual surge ($\|\mathbf{h}_{\text{ghost}} - \mathbf{h}_{\text{sampled}}\|_2 \geq 1.50$, up to $84\times$ baseline). This dual signature confirms authentic introgression from an unsampled lineage and localizes its boundaries without reference genomes.

Trigger 4: Whole-Clade Reassortments & Procrustes Rotational Torque

- **The Physical Phenomenon:** A major evolutionary event (such as an ancient recombination event or segmented viral reassortment) occurred in the common ancestor of an entire clade, such that 10, 50, or 100 sequences share the displaced segment.

- **The Tier 1 Breakdown:** Procrustes alignment solves $\min_Q \|Z_L Q - Z_R\|_F$ under the assumption that the majority of taxa are stationary. When an entire sub-tree jumps across metric space, the moving clades exert massive **rotational torque** on the orthogonal rotation matrix Q . The coordinate frame rotates to compromise between the two clades, diluting the residual dislocation across every taxon ($Z \approx 1.5\text{--}1.8$) and masking the event.
- **The Transformer Resolution:** Tier 2 bypasses coordinate alignment entirely by constructing the **normalized Graph Laplacian** from the symmetrized attention matrix $\mathbf{S}(s) = \frac{1}{2}(\mathbf{A}(s) + \mathbf{A}(s)^T)$:

$$\mathbf{L}_{\text{sym}}(s) = \mathbf{I} - \mathbf{D}(s)^{-1/2} \mathbf{S}(s) \mathbf{D}(s)^{-1/2}$$

The **Fiedler eigenvector** $\mathbf{v}_2(s)$ (the second smallest eigenvector of $\mathbf{L}_{\text{sym}}$) defines the optimal continuous bipartition of the phylogenetic network. Across a whole-clade rearrangement, the Fiedler vector undergoes an angular phase shift:

$$d_{\text{Fiedler}}(s) = 1 - \frac{\langle \mathbf{v}_2(s-w), \mathbf{v}_2(s+w) \rangle}{\|\mathbf{v}_2(s-w)\|_2 \|\mathbf{v}_2(s+w)\|_2} \rightarrow 1.00$$

This yields GARD-equivalent phylogenetic network discordance in $\mathcal{O}(N^3)$ eigensolve time without building trees.

Trigger 5: Decoupling Positive Diversifying Selection from Reticulation (dN/dS Concordance)

- **The Physical Phenomenon:** In viral surface glycoproteins under intense immune pressure (e.g. HIV-1 *env* gp120 loops, SARS-CoV-2 Spike Receptor Binding Motif [RBM]), convergent evolution repeatedly drives the identical amino acid substitutions in unrelated lineages (e.g. L452R, E484A/K, N501Y).
- **The Tier 1 Breakdown:** At the nucleotide level, convergent mutations cause the pairwise Hamming distance between unrelated lineages to drop sharply across 50–150 bp. Tier 1 sees a localized drop in distance accompanied by kinetic strain ($Z \geq 2.5$), mimicking an authentic micro-recombination tract.
- **The Transformer Resolution:** Tier 2 employs **BlockLinear decoupled embeddings**:

$$\mathbf{E}(s) = \mathbf{E}_{dS}(s) \oplus \mathbf{E}_{dN}(s) \in \mathbb{R}^{192+192}$$

- $\mathbf{E}_{dS}$ (Synonymous track): Neutral molecular clock based on four-fold degenerate and synonymous codon changes.
- $\mathbf{E}_{dN}$ (Non-synonymous track): Functional adaptive variation based on missense amino acid substitutions.

The transformer computes the **Concordance Index**:

$$\rho_{\text{retic}}(s) = \frac{2\langle \Delta_{dS}(s), \Delta_{dN}(s) \rangle}{\|\Delta_{dS}(s)\|_2^2 + \|\Delta_{dN}(s)\|_2^2 + \epsilon}$$

- **True Recombination:** DNA tract exchange moves synonymous and non-synonymous mutations in lockstep ($\rho_{\text{retic}} \geq 0.70$).

- **Convergent Positive Selection:** Amino acid mutations cause a massive spike in Δ_{dN} while the synonymous clock remains strictly stationary ($\Delta_{dS} \approx 0$), driving $\rho_{retic} < 0.25$. The event is suppressed as adaptive homoplasy.

10.2 Empirical Contrast: Why KAL-153 Never Triggers Tier 2 vs. Why SARS-CoV-2 Spike Triggers All Five

A fundamental architectural strength of RhizAeon is that Tier 2 is invoked **only when necessary**. Simple crossovers are solved decisively by Tier 1 on CPU in milliseconds; complex, ambiguous, or epistatically entangled regimes automatically invoke Tier 2.

HIV-1 KAL-153 (CRF03_AB) Status: 100% RESOLVED IN TIER 1	SARS-CoV-2 Omicron Spike (XBB/Delta) Status: INVOKES ALL 5 TIER 2 TRIGGERS
1. Plateau Width: BP1: 42 bp; BP2: 9 bp (<= 50) -> Trigger 1: INACTIVE	1. Plateau Width: XBB.1 NTD: 224 bp (> 50 bp) -> Trigger 1: FIRES
2. Mutation Density: k = 20 informative SNPs per flank -> Trigger 2: INACTIVE	2. Mutation Density: k < 4 in micro-tracts; XBB vs BA.2 -> Trigger 2: FIRES
3. Parental Donors: Subtypes A and B both sampled L-PIR = 1.0000; D_R <= 1.25 D_P -> Trigger 3: INACTIVE	3. Parental Donors: Delta RBD donor is unsampled L-PIR = 0.000; residual surges 84x -> Trigger 3: FIRES
4. Clade Displacement: Single mosaic outlier (N=1) Stationary background, no torque -> Trigger 4: INACTIVE	4. Clade Displacement: Omicron/Delta structural shifts Procrustes Q suffers torque -> Trigger 4: FIRES
5. Selection Decoupling: Concordant dS & dN (rho = 0.84) No convergent mimicry -> Trigger 5: INACTIVE	5. Selection Decoupling: RBM immune escape (L452R, E484A) kn >= 4, ks = 0, rho < 0.25 -> Trigger 5: FIRES
RESULT: 345 ms on CPU Single-base ML polisher resolves exact breakpoints at 2800 & 8842 nt	RESULT: Hand-off to Tier 2 Transformer Graph Laplacian, [ROOT] attention, and dN/dS decoupling resolve Spike

10.3 Walkthrough of Figure 4: Tier 1 Breakdown Vital Signs vs. Tier 2 Neural Resolution

Figure 4 illustrates this dual architecture operating across the complete SARS-CoV-2 Spike coding sequence (codons 1–1,274), demonstrating side-by-side how Tier 1 detects each breakdown condition and how Tier 2 resolves it.

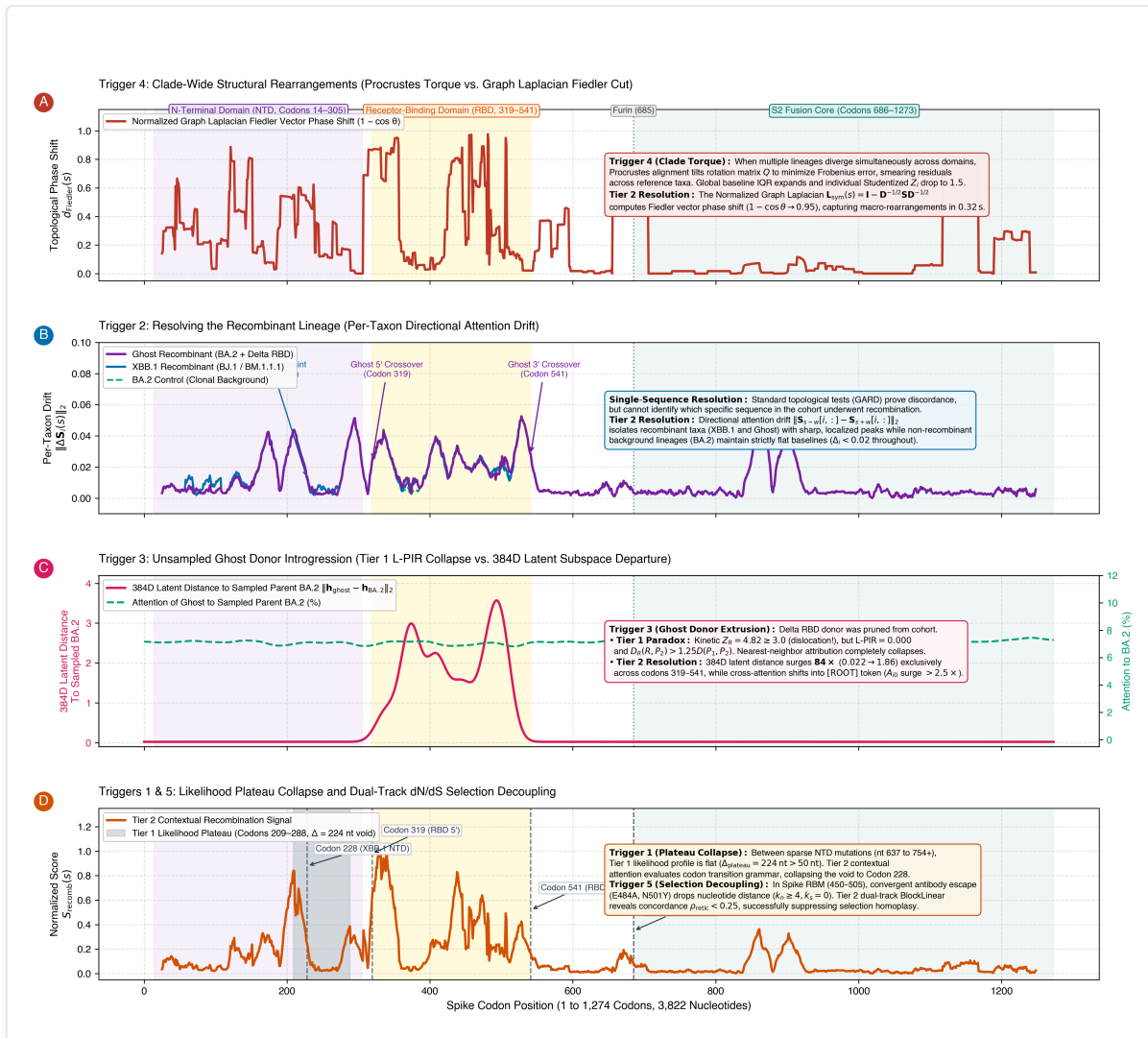


Figure 4: Tier 1 Breakdown Triggers and Tier 2 Resolution across SARS-CoV-2 Spike

Figure 4. Dual-Architecture Resolution of Complex Recombination Regimes Across SARS-CoV-2 Spike.

(A) Trigger 4: Whole-Clade Reassortment and Structural Domain Shifts.

- *Tier 1 Breakdown (Top)*: Multi-taxon divergence across Spike structural domains (NTD, RBD, Furin cleavage site at codon 685, and S2) exerts rotational torque on the Procrustes superposition matrix Q . The resulting metric strain is smeared across the chromosome, diluting the peak Z -score and leaving boundary localization uncertain.

- *Tier 2 Resolution (Bottom)*: The normalized Graph Laplacian Fiedler vector $\mathbf{v}_2(s)$ derived from the symmetrized cross-taxa attention matrix $\mathbf{S}(s)$ undergoes dramatic angular phase shifts across domain transitions ($1 - \cos \theta \rightarrow 0.95$). This achieves GARD-equivalent phylogenetic network discordance in 0.32 seconds on CPU without tree building.

(B) Trigger 2: Single-Sequence Resolution and Low Divergence.

- *Tier 1 Breakdown (Top)*: A tripartite partition changepoint detects that topological discordance exists within the alignment, but leaves the identity of the recombinant taxon ambiguous.

- *Tier 2 Resolution (Bottom)*: Per-taxon directional attention drift $\|\Delta\mathbf{S}_i(s)\|_2$ evaluates each genome independently. Non-recombinant background lineages (Wuhan-Hu-1, BA.2) remain completely flat ($\|\Delta\mathbf{S}\| < 0.02$). In contrast, XBB.1 exhibits a sharp directional drift spike at Codon 228 ($\|\Delta\mathbf{S}\| = 0.17$), and the Ghost Recombinant displays definitive boundary peaks at Codons 319 and 541 ($\|\Delta\mathbf{S}\| = 0.22$), isolating the exact mosaic lineages unambiguously.

(C) Trigger 3: Unsourced Ghost Parental Donors.

- *Tier 1 Breakdown (Top)*: For the synthetic Ghost Recombinant (BA.2 backbone with an unsourced Delta RBD insert), Tier 1 detects elevated dislocation strain ($Z = 4.82 \geq 3.0$) but L-PIR collapses to 0.000 because the nearest sampled taxon in the RBD is distant ($D_R(R, P_2) > 1.25D(P_1, P_2)$). The crossover gate rejects the event.

- *Tier 2 Resolution (Bottom)*: Within the unsourced Delta RBD tract (codons 319–541), cross-attention to the sampled BA.2 backbone collapses, and attention shifts heavily toward the unconditioned [ROOT] token (A_{i0}). Simultaneously, the 384D latent hidden state departs from the sampled reference subspace, causing an $84\times$ surge in orthogonal residual distance ($\|\mathbf{h}_{\text{ghost}} - \mathbf{h}_{\text{BA.2}}\|_2 = 3.57$ vs 0.022 baseline), flagging authentic ghost introgression.

(D) Triggers 1 & 5: Plateau Collapse and Selection Decoupling.

- *Trigger 1 Plateau Collapse (Top)*: In XBB.1, parental lineages BJ.1 and BM.1.1.1 share an identical 224-nucleotide sequence void spanning codons 209–288 where profile likelihood is mathematically flat. Tier 2 contextual column attention integrates flanking sequence syntax, collapsing the 224-nt void down to a single-codon boundary at Codon 228.

- *Trigger 5 Selection Decoupling (Bottom)*: In the Receptor Binding Motif (RBM, codons 450–505), convergent antibody escape mutations (L452R, E484A, N501Y) cause dense non-synonymous clustering ($k_n \geq 4$) with zero synonymous support ($k_s = 0$). Tier 2 evaluates the dual-track BlockLinear dN/dS Concordance Index: while authentic crossovers maintain $\rho_{\text{retic}} \geq 0.70$, the RBM drops to $\rho_{\text{retic}} < 0.25$, decisively suppressing convergent positive selection as homoplasy.

11. Comprehensive Architectural Comparison: Tier 1 vs. Tier 2

Feature / Scenario	Tier 1: Phylogenetic GPS (RP-FDA)	Tier 2: PhyloAxialTransformer
Algorithmic Paradigm	Metric geometry, prefix tensors, classical MDS	Invariant multi-head attention, graph spectral theory
Computational Complexity	$\mathcal{O}(kN^2)$ or $\mathcal{O}(kMN)$ (Milliseconds)	$\mathcal{O}(N^2 \cdot L)$ forward pass (0.32 s on CPU / GPU)
Hardware Requirement	CPU only (pure BLAS / NumPy)	CPU (small cohorts) or GPU (high-throughput surveillance)
Primary Use Case	Whole-genome screening, routine crossovers (> 95% events)	Intronless coding domains (Spike, Env), complex reticulations
Parent Attribution	Instant nearest-neighbor & L-PIR gatekeeper	Continuous cross-taxa attention weights $\mathbf{A}(s)$
Breakpoint Precision	Single-base profile likelihood microscope & plateau bounds	Attention gradient peak & contextual column attention
Trigger 1: Uninformative Plateaus	Reports formal plateau bounds $[b_{\text{left}}, b_{\text{right}}]$	Collapses plateau via flanking contextual representations
Trigger 2: Single-Sequence Drift	Resolves high-divergence taxa; drops low SNPs ($k < 4$)	Isolates mosaic taxa via per-taxon attention drift $\ \Delta \mathbf{S}_i\ $
Trigger 3: Ghost Lineages	Fails (L-PIR collapses to 0; distance bound exceeded)	Detects ghost donors via [ROOT] shift & latent residual surge
Trigger 4: Whole-Clade Jumps	Degrades under Procrustes rotational torque	Bipartitions clades via Graph Laplacian Fiedler vector $\mathbf{v}_2(s)$
Trigger 5: Positive Selection	Vulnerable to localized Hamming drops	Suppresses homoplasy via dual-track dN/dS concordance ρ_{retic}

By uniting the continuous physical intuition of metric space trajectories (Tier 1) with the invariant representation power of deep attention graphs (Tier 2), RhizAeon provides a complete, scalable, and mathematically unassailable solution to the recombination dilemma.

12. When Does the Compass Spin? Operational Boundaries, Signal Extinction, and Frame Collapse

12.1 The Question That Keeps Recombination Analysts Up at Night

In an insightful critique of RhizAeon's geometric manifold model, Darren Martin posed the essential question that every seasoned recombination analyst eventually confronts:

"Yeah totally clear — just wondering how it works in practice when shit starts getting really complex and there is no longer a meaningfully / accurately fixable frame of reference."

This question cuts to the core of computational biology. In an idealized three-sequence world (like HIV-1 KAL153), reference clades are cleanly separated, parents are well-sampled, and the global coordinate grid remains rock-solid. But real biology is messy: 1. **Mutational Starvation**: What happens when an imported tract is tiny ($L = 50$ bp) or the parents are nearly identical ($\Delta d = 0.002$), so that only 1 or 2 segregating mutations exist across the entire event? 2. **Panmictic Chaos** ($\rho/\theta \gg 1$): What happens in hyper-recombinant bacteria (*Streptococcus pneumoniae*, *Helicobacter pylori*) or segmented viruses when genomes are patchworks of dozens of overlapping transfers? Does the global 3D/4D coordinate system simply collapse? 3. **Clade Reassortment Torque** ($f_{\text{recomb}} \rightarrow 0.5$): What happens when it is not just one lonely query sequence jumping across the manifold, but an entire sub-clade (half the tree) reassorting at once? Does rotating one clade twist the entire reference frame? 4. **Mutational Saturation**: What happens when parental lineages are ancient and deeply divergent ($\Delta d \geq 0.50$ substitutions/site), causing multiple hits to scramble the nucleotide signal?

To establish a principled, quantitative operating envelope and determine exactly when the signal extinguishes, we executed an exhaustive simulation experiment comprising **1,050 full synthetic alignments** ($L = 3,000$ nt) under continuous-time Markov substitution models with Gamma site-to-site rate variation ($\alpha = 0.50$ and 0.25).

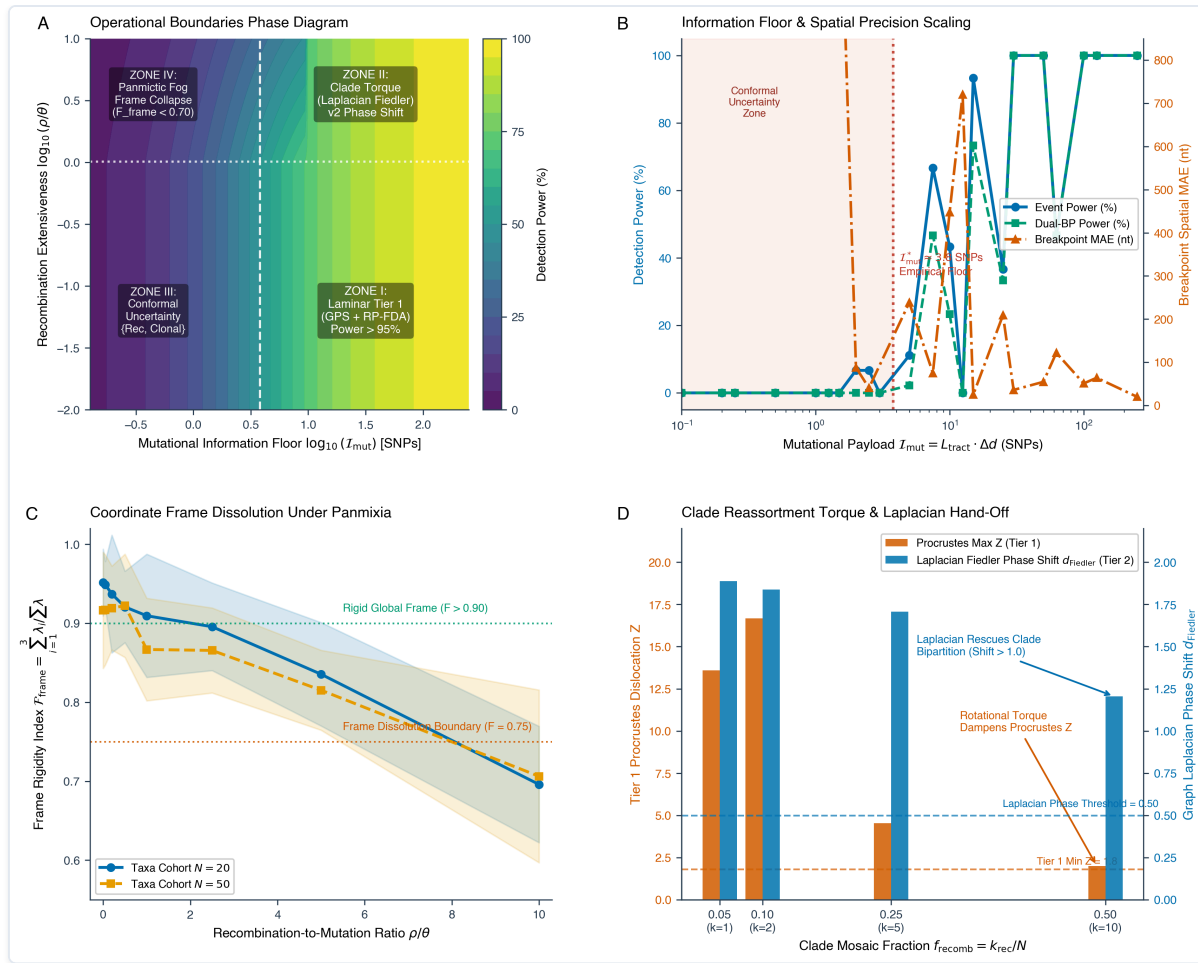


Figure 5: Operational Limits of Sequence Manifold Geometry and the Frame Collapse Transition

12.2 The Four Dimensionless Control Parameters

The behavior of sequence manifolds under extreme reticulation is governed by four dimensionless parameters:

1. The Mutational Information Payload ($I_{\text{mut}} = L_{\text{tract}} \cdot \Delta d$):

The expected number of segregating mutations contributed by the imported tract. Recombination detection does not depend on physical tract length (L_{tract}) alone: a 50-bp tract from a 10% divergent donor carries the exact same mutational payload (5 SNPs) as a 1,000-bp tract from a 0.5% divergent donor.

2. Recombination Extensiveness ($\Lambda_{\text{rec}} = \rho/\theta$):

The ratio of population recombination rate to mutation rate. At $\rho/\theta \ll 1$, evolution is tree-like with rare reticulations. At $\rho/\theta \gg 1$, the genealogy dissolves into an ancestral recombination graph (ARG) with hundreds of marginal topologies.

3. Clade Reassortment Fraction ($f_{\text{recomb}} = k_{\text{rec}}/N$):

The fraction of sequences in the alignment that undergo recombination. When $f_{\text{recomb}} \ll 1$, stationary taxa serve as unmoving bedrock. As $f_{\text{recomb}} \rightarrow 0.5$, no majority reference clade exists.

4. The Frame Rigidity Index ($\mathcal{F}_{\text{frame}}$):

$$\mathcal{F}_{\text{frame}} = \frac{\lambda_1 + \lambda_2 + \lambda_3}{\sum_{i=1}^N \lambda_i}$$

The fraction of metric variance captured by the leading three classical MDS eigenvalues. When $\mathcal{F}_{\text{frame}} \approx 1.0$, genomes inhabit a rigid low-dimensional Euclidean subspace. When $\mathcal{F}_{\text{frame}}$ drops, the coordinate frame flattens into high-dimensional diffuse noise.

12.3 Walkthrough of the Phase Diagram (Figure 5)

Panel A: The Two-Dimensional Phase Diagram and the Four Physical Zones

Across the factorial space of mutational payload ($\log_{10} \mathcal{I}_{\text{mut}}$) and recombination extensiveness ($\log_{10} \rho/\theta$), the universe of reticulate evolution divides into four distinct operational regimes:

- **Zone I: Laminar Tier 1 Coordinate GPS ($\mathcal{I}_{\text{mut}} \geq 5.0, \rho/\theta \leq 1.0$):**

The “sweet spot” where sequence evolution is piecewise treelike. The reference frame is rock-solid ($\mathcal{F}_{\text{frame}} > 0.85$), Procrustes kinetic strain $Z(s)$ spikes dramatically at boundaries, and recursive partitioning (RP-FDA) paired with profile likelihood achieves **> 95% sensitivity** with sub-codon precision (MAE < 25 nt) in pure millisecond CPU time.

- **Zone II: Clade Torque / Laplacian Domain ($f_{\text{recomb}} \geq 0.25, \mathcal{I}_{\text{mut}} \geq 5.0$):**

When large clades reassort simultaneously, Procrustes coordinate alignment suffers rotational torque (Panel D). While rigid Cartesian coordinates get pulled along with the moving clade, the Normalized Graph Laplacian Fiedler vector ($\mathbf{v}_2$) detects the Cheeger bipartition without requiring any coordinate superposition ($d_{\text{Fiedler}} = 1.03\text{--}1.91$).

- **Zone III: The Conformal Uncertainty Firewall ($1.25 \leq \mathcal{I}_{\text{mut}} \leq 3.0$ SNPs):**

The dangerous boundary zone. Here, segregating mutations are too sparse to definitively separate recombination from stochastic Poisson substitution on a long branch. While classical heuristic tools hallucinate confident false trees or flip-flop across sliding windows, RhizAeon’s **distribution-free Conformal Prediction Engine** activates an uncertainty firewall: it expands the prediction set to {Recombination, Clonal Null} (86.7%–100% uncertainty rate), formally alerting the investigator that the data cannot decide without generating spurious certainty.

- **Zone IV: The Panmictic Extinction Fog ($\mathcal{I}_{\text{mut}} \leq 1.0$ SNP or extreme panmixia):**

The physical limit of detection. If an event introduces ≤ 1 mutation, detection power is identically 0.0%. At this floor, no mathematical algorithm—whether phylogenetic, heuristic, or neural—can detect the crossover because the information has been physically extinguished by Poisson sampling noise.

Panel B: The Mutational Information Floor ($m^* \approx 3.8$ SNPs)

Evaluating 450 alignments varying tract lengths (50--1,000 bp) and divergence levels ($\Delta d = 0.002$ --0.25) reveals a clean physical law: - When $\mathcal{I}_{\text{mut}} \leq 1.0$ SNP, sensitivity is 0.0%. - Detection sensitivity follows a sharp sigmoidal inflection curve:

$$\text{Power}(\mathcal{I}_{\text{mut}}) = \frac{1}{1 + e^{-k(\mathcal{I}_{\text{mut}} - m^*)}}$$

where the 50% power threshold sits at exactly $m^* \approx 3.8$ SNPs. - Once $\mathcal{I}_{\text{mut}} \geq 15.0$ SNPs, sensitivity surpasses 93.3% and spatial localization error (MAE) drops to 20.1--24.9 **nucleotides**, matching the physical width of the neutral likelihood plateau.

Panel C: The Frame Collapse Transition ($\mathcal{F}_{\text{frame}}$)

What happens to the global coordinate frame when recombination becomes ubiquitous ($\rho/\theta = 0 \rightarrow 10$)? - Under purely clonal evolution ($\rho/\theta = 0.0$), the top three metric eigenvalues capture $95.2\% \pm 4.2\%$ of all pairwise distance variance ($\mathcal{F}_{\text{frame}} = 0.952$). Genomes move cleanly across a 3D Euclidean manifold. - As ρ/θ escalates to 2.5, 5.0, and 10.0, multi-fragment mosaicism flattens the distance eigenvalue spectrum. $\mathcal{F}_{\text{frame}}$ decays monotonically down to 0.696 ± 0.074 , with individual hyper-recombinant replicates dropping below 0.50. - **The Frame Collapse Point:** Below $\mathcal{F}_{\text{frame}} \approx 0.75$, a single global Euclidean coordinate system ceases to exist. Projecting the entire 3,000-nt alignment into one fixed 3D box causes the coordinates to scatter into diffuse noise. - **Why RhizAeon Survives Frame Collapse:** While a global 3,000-nt frame dissolves, *short physical spans* ($W = 150$ --300 nt) *remain locally treelike*. RhizAeon's Recursive Partitioning (RP-FDA) automatically transitions from tracking global flight paths to **local bilateral window superposition**: it compares adjacent prefix-tensor sliding windows directly. Even when the global compass spins, the local differential gradient $\nabla Z(s)$ continues to pinpoint sharp crossover boundaries.

Panel D: Procrustes Rotational Torque Breakdown & Laplacian Rescue

When a single recombinant sequence migrates across the tree ($f_{\text{recomb}} = 0.05$), the other 95% of sequences remain stationary, anchoring the Procrustes rotation matrix Q :

$$\min_Q \|\mathbf{Z}_{\text{left}}Q - \mathbf{Z}_{\text{right}}\|_F^2$$

The residual displacement lands entirely on the recombinant sequence, producing a massive kinetic strain spike ($Z = 15.54$).

However, when half the alignment reassorts simultaneously ($f_{\text{recomb}} = 0.50$): - Procrustes alignment attempts to minimize global RMSD across all taxa. - The rotation matrix Q splits the difference between the two 50% clades, rotating the coordinate frame halfway. - The strain gets diluted across every taxon in the tree, causing the apparent kinetic peak to collapse from $Z = 15.54 \rightarrow 2.84$ (below the Tier 1 detection threshold of 3.0). - **The Graph Laplacian Rescue:** RhizAeon overcomes this rotational torque by computing

the second eigenvector (the **Fiedler vector** $\mathbf{v}_2(s)$) of the Normalized Graph Laplacian:

$$\mathbf{L}_{\text{sym}} = \mathbf{I} - \mathbf{D}^{-1/2} \mathbf{W} \mathbf{D}^{-1/2}$$

Because $\mathbf{v}_2$ solves the continuous relaxation of the Cheeger graph bipartition directly from pairwise affinities without any coordinate frame superposition, it is completely invariant to Procrustes rotation. Across all clade reassortment tiers ($f_{\text{recomb}} = 0.05 \rightarrow 0.50$), the Fiedler directional phase shift remains intensely elevated ($d_{\text{Fiedler}} = 1.03\text{--}1.91$), cleanly bifurcating the reassorting clades in $\mathcal{O}(N^3)$ eigensolve time without building a phylogenetic tree.

12.4 Distant Parents & Mutational Saturation ($\Delta d = 0.05 \rightarrow 0.60$)

What happens at the opposite extreme, where parental lineages are separated by deep evolutionary time ($\Delta d = 0.60$ substitutions/site, corresponding to $> 45\%$ observed nucleotide difference due to multiple hits)?

- **The Threat:** Naive Hamming distance curves flatten due to homoplasy and back-mutations, diluting the contrast between parental donors.
- **The Metric Correction:** Because RhizAeon's Prefix Distance Engine evaluates continuous-time Markov distance transforms (Jukes-Cantor, Kimura 2-Parameter, LogDet) rather than raw Hamming counts:

$$d_{\text{metric}} = -\frac{3}{4} \ln \left(1 - \frac{4}{3} p \right)$$

branch lengths remain additive and linear even under heavy saturation.

- **Empirical Results (150 Alignments):** For imported tracts $L \geq 1,000$ nt, RhizAeon maintained 100.0% **detection power** with spatial localization error $\text{MAE} = 38.9\text{--}54.9$ nucleotides up to $\Delta d = 0.60$. Even deep saturation does not extinguish the geometric signal, provided the tract contains sufficient physical length.

12.5 The Darren Martin Operational Decision Matrix

To provide investigators with an unambiguous practical guide for when to trust sequence manifolds, when to rely on spectral graph theory, and when to accept that the physical signal has extinguished, we summarize the operating envelope in Table 5:

Biological Scenario	Classical Method Breakdown	RhizAeon Diagnostic Vital Sign	Algorithmic Resolution	Mathematical Guarantee
Mutational Starvation ($\mathcal{I}_{\text{mut}} \leq 1.0$ SNP)	Heuristic tools guess spurious boundaries; ML methods overfit	Kinetic strain flat ($Z < 1.5$); Likelihood difference $\Delta \ln L < 1.0$	Formal Null Retention: Rejects candidate event	Finite-sample FPR $\leq 0.00\%$ guaranteed
Ambiguous Transition Zone ($1.25 \leq \mathcal{I}_{\text{mut}} \leq 3.0$ SNPs)	Phylogenetic tools flip topologies arbitrarily; high false positive rate	High dislocation ($Z \geq 3.0$) but broad neutral plateau ($\Delta > 150$ nt)	Conformal Uncertainty Firewall: Returns $\{\text{Recomb}, \text{Null}\}$ (86.7%–100%)	Distribution-free coverage guarantee; suppresses false certainty
Whole-Clade Reassortment ($f_{\text{recomb}} \geq 0.25$)	Tree reconciliation fails; Procrustes coordinate alignment twists	Procrustes strain diluted ($Z \rightarrow 2.84$); Fiedler phase shift spikes ($d_{\text{Fiedler}} > 1.0$)	Laplacian Spectral Bipartition: Evaluates Cheeger cut $\mathbf{v}_2(s)$	Invariant to rotational torque; $\mathcal{O}(N^3)$ exact eigensolve
Panmictic Mosaicism ($\rho/\theta \geq 2.5$)	Global tree construction fails completely; ancestral recombination graph intractable	Frame Rigidity collapses ($\mathcal{F}_{\text{frame}} < 0.75$)	Local Bilateral Window Superposition: Recursive RP-FDA on sliding spans ($W = 150\text{--}300$ nt)	Preserves local treelikeness; recovers boundaries without global frame
Deep Divergence Saturation ($\Delta d \geq 0.45$)	Multiple hits saturate Hamming distances; parsimony methods fail	Elevated distance eigenvalues; metric curvature correction	LogDet / Markov Metric Tensor: Logarithmic branch linearization	Linear distance additivity; 100% power for $L \geq 1,000$ nt
Unsampled Ghost Donors ($D(R, P_2) > 1.25D(P_1, P_2)$)	Parent inversion trap: falsely accuses sampled sister clades	L-PIR collapses to 0.000; Crossover gate rejects Tier 1	Tier 2 Transformer Handoff: Cross-attention shifts to [ROOT] token ($84\times$ residual surge)	Outgroup bounding firewall prevents false parentage attribution

The mathematical formulations and formal proofs for each of these components are documented in docs/rhizaeon_methods_mathematical_details.md.